

DYNAMICMS SUITE

Dynamic Apps Standard SRS

Common Software Requirements Specification for DynamicMS Suite Applications

Consolidated from journi (DynamicJourni) v2.0 and NCP Solver v2.0, extended from the CortexPLM reference build

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) — “Dynamic Apps Standard SRS” — defines the set of functional and non-functional requirements that are common to every application in the DynamicMS Suite, irrespective of the business domain a given application addresses. It is derived by comparing the individual SRS documents of two suite members — *journi* (DynamicJourni), a human change-management application, and NCP Solver, a non-conformity and problem-resolution application — and extracting the requirements, architecture patterns, and governance mechanisms that recur identically, or near-identically, across both.

This document is intended to serve as the baseline specification a new Dynamic App is built against, and as the reference against which an existing Dynamic App's platform layer (as distinct from its domain-specific modules) is assessed for conformance. It is not a replacement for either product-specific SRS: *journi* and NCP Solver each retain their own SRS for domain-specific functional requirements (the 22 change-management modules on one side, the S1–S7 NCP Sheet workflow on the other).

Revision 1.1 extends the standard with platform capabilities first built and verified in CortexPLM, an innovation-lifecycle Dynamic App, where they proved independent of that application's business domain — for example the pinnable, repositionable navigation menu, internal benchmarking, and in-place upgrade and backup. Only requirements applicable to any Dynamic App were retained; CortexPLM's phase-gate lifecycle and its product-specific content remain in its own documentation. Appendix B lists every change made in this revision.

Revision 1.2 adds a second set of capabilities proven in CortexPLM that any Dynamic App can reuse: the tenancy tree (Group Yes/No > Organization > principal records) and the provisioning of new organizations, the portfolio overview matrix, a stage checklist library, multi-format attachments, AI shown and run where it applies in the process design, the choice of the live AI model, a full-screen BPMN workspace, on-screen legends and a decision matrix for scored fields. Appendix C lists these changes.

Revision 1.3 integrates the application-agnostic requirements for the Vertical and SME layers — the vertical framework, vertical processes, data, compliance, integration and configuration, the SME mode with its tracks, complexity scoring, onboarding, packaging and pricing, and their combination — restated as Dynamic App requirements. It also adds, for any Dynamic App, a project template catalog (universal or per vertical, Full or SME mode) with full CRUD, three ways to create a project (from the catalog, manually, or with AI), a Checklist Library whose checklists may or may not be related to a gate, and a Gate Library to which checklists are associated; both are managed with full CRUD and can be attached to any project phase. Appendix D lists the changes and Appendix E traces every source requirement family to its requirements here.

Revision 1.4 adds questionnaire and survey management, first specified for CortexSkills (change request CR1, Scope of Work Definition & Questionnaire Management) and restated as a capability any Dynamic App can reuse: versioned questionnaire templates tagged by focus, sector and language; elaboration by loading a template, by AI with mandatory human validation, or manually; RACSI allocation per section; distribution through Face-to-Face, Email, Hybrid and Custom channels with

per-channel reminders; consent-gated, offline-capable response collection; and consolidation of complete responses into a tenant-scoped dataset that feeds the application's downstream analysis.

Revision 1.5 adds capabilities first built and verified in DynamicMS, an integrated management system Dynamic App, together with product direction received with its user feedback, restated so that any Dynamic App can reuse them: a global search reached from the navigation menu; full Create, Read, Update and Delete, with version history and the restore of any version, on every element of the process design (functions, end-to-end processes, phases, macro processes, tasks, steps, gates, checklists, forms and templates); an OBS made of units, roles and people, in which a role is linked to one or more functions and a person plays one or more roles; AI Use Cases mapped to the exact task or step they support, each with a prompt specification whose aspects (role, context, task, constraints, examples, format and others) are separate, versioned fields fully populated for that task or step; explicit naming and descriptions of process elements; typed step forms and a readiness checklist before a process starts; documented-information templates filled from the application's own data; versioned attachments; and an Organization-level language-model configuration. Appendix G lists every change.

Revision 1.6 adds what the second and third rounds of DynamicMS user feedback established, restated for any Dynamic App: generated documents that are consistent with the application's data and with each other, detailed and fine grained by document type, with full Create, Read, Update and Delete at every level (document, version, section, data-bound row, template and layout); data entry that is usable for real work (qualified field titles, controlled lists with a Custom value, a large row editor, scrollable dialogs, identifiers shown with their names, a search bar on every screen); records linked many to many, entered manually, from a library or by the AI, and propagated to registers and documents; graded audit and inspection findings; and transparent AI behavior (which engine answered and why, parameters adapted to each model). Appendix H lists every change.

1.2 Scope

In scope: the tenant and organizational hierarchy; identity and role-based access control; Configuration Management (Solution Packs, Add-Ons, Compliance & Security Standards); the governed AI Use Case Library; per-process/per-stage governance artifacts (Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, RACSI accountability); a Standards/Knowledge Base with RAG grounding; BPMN 2.0 process modeling; standard reports and data export; an in-app Notification Center; an AI Assistant / chatbot offering data-query and application-help modes; an audited, justification-governed change log; an in-app Help guide; template libraries; multi-language localization including Arabic RTL; and the non-functional requirements (performance, security, reliability, usability, portability, maintainability, compatibility, data retention) these capabilities are held to.

Revision 1.1 adds to this scope: a personalizable navigation shell (a menu pinned open or sliding away, docked on any edge of the screen, with favorites that follow the user); internal benchmarking within an Organization and across the Organizations of one Group; and platform operations (scheduled backup with retention, additive startup migrations, startup guards, a health endpoint, a reproducible demonstration seed, and continuous integration across operating systems).

Revision 1.4 adds to this scope: questionnaire and survey management — questionnaire templates and their library, questionnaire elaboration, validation and RACSI allocation, multi-channel distribution and reminders, consent-based and offline response collection, completeness validation, consolidation and response reporting.

Revision 1.5 adds to this scope: global search; process design management with version history and restore; the OBS role model (functions, roles and people); AI Use Case mapping and prompt specification; process element naming and descriptions; typed step forms and process readiness; documented-information templates and generated documents; versioned attachments.

Revision 1.6 adds to this scope: consistency and fine-grained content of generated documents; full CRUD on documents, their versions, sections and data-bound content; data entry usability; linked records, choice libraries and register propagation; audit and inspection programmes with graded findings; AI engine transparency and model capability profiles.

Out of scope: any requirement specific to a single application's business domain — journi's 22 change-management modules and their ADKAR/Lewin/Bridges/Kotter/Kübler-Ross framework content, or NCP Solver's seven-stage NCP Sheet lifecycle and its Ishikawa/5-Why root-cause tooling. Those remain the authority of their respective product SRS documents.

Also out of scope: external benchmarking — any comparison with an organization outside the requester's Group, or with third-party or industry data.

1.3 Intended Audience

Product owners defining a new Dynamic App; architects and developers building or rebuilding a Dynamic App's platform layer; quality assurance engineers verifying platform-level conformance; and auditors assessing governance, security, and compliance posture across the suite.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Dynamic App	Any application built on the DynamicMS Suite platform pattern described in this document (e.g. journi / DynamicJourni, NCP Solver).
Solution Pack	One of three commercial tiers defining an Organization's feature and quota entitlement matrix.
Add-On	A priced, independently toggleable capability that grants a feature to a Pack not already including it standard.
Compliance & Security Standard	An independently priced/toggleable module (e.g. GDPR, ISO/IEC 27001, SOC 2) that seeds GRC scaffolding; explicitly not a certification.
AI Use Case	A governed, versioned catalog entry describing one AI-assisted capability, tiered Assistive or Augmented (never Autonomous).
RAG	Retrieval-Augmented Generation — search over a knowledge corpus (TF-IDF/cosine-similarity or equivalent) grounding an AI suggestion or chatbot answer in the tenant's own data or a static reference corpus, without requiring a live external LLM call.
OBS	Organizational Breakdown Structure — the site/department/team or resourcing tree other governance modules reference for ownership and assignment.
RACSI	Responsible, Accountable, Consulted, Support, Informed — a five-role accountability model; exactly one Accountable per activity.
COSO	Committee of Sponsoring Organizations of the Treadway Commission — the internal-control framework used to classify Controls.
BPMN	Business Process Model and Notation — the OMG 2.0 standard used to model process workflows in-browser.
RBAC	Role-Based Access Control — access rights determined by one or more assigned roles, mediated by a runtime-editable Permission Matrix.

Term	Definition
Justification Governance	The stage-then-justify pattern: a practitioner stages a new value, records a justification note, and only then commits the change, appended atomically to an audit/change log.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with isolated data.
Work Breakdown Structure (WBS)	A hierarchical decomposition of any grouping of Actions into a tree of WBS nodes, used as the basis for Gantt scheduling.
Gantt Chart	A timeline rendering of a WBS, plotting each node/Action's planned dates and dependencies.
Version Management	The platform's generic service for versioning any registered entity type: immutable history, no destructive edits, and revert-by-duplication to a new Current version.
Multi-Channel Communication	Delivery of a notification or alert through one or more channels (in-app, email, SMS, push, webhook) behind a single shared dispatch interface.
Mobile Compatibility	A responsive layout, meeting WCAG 2.1 AA touch-target sizing, that keeps an application's core workflows fully usable on a mobile viewport without a separate native application.
Return on Experience (REX)	A structured lessons-learned entry captured at the closure of a Project or an Action — what went well, what didn't, root cause, recommendation — indexed for future retrieval.
External Integration	A registered, independently toggleable connection from the platform to an external application, configured through the External Integration Registry (Section 3.3).
Navigation Shell	The frame shared by every authenticated screen: the primary menu, the header bar (context switcher, language, notifications, AI Assistant, user menu), and the content area.
Pinned / Sliding Menu	The two menu modes. Pinned, the menu stays open and the content reflows beside it. Unpinned, it collapses to an edge handle and slides over the content when pointed at, focused, or toggled.
Dock Position	The screen edge — left, right, top, or bottom — a user chooses for the primary menu; a per-user preference stored server-side.
Internal Benchmarking	Comparison of standard metrics between segments of one Organization, or between Organizations of the same Group, using aggregates only. External benchmarking is out of scope.
Minimum Sample Size	The smallest number of records a benchmark segment or Organization must contain to be ranked; smaller samples are shown as not comparable.
Additive Migration	A change applied at startup that only adds tables, columns, permission codes, or default grants, and never drops or rewrites existing data or an administrator's edits.
Tenancy Tree	The view Group (Yes/No) > Organization > principal records, limited to what the viewer may see; an Organization with no Group appears as Independent.
Provisioning	Everything a new Organization receives on creation: OBS skeleton, standard governance catalog, AI Use Case Library, checklist and record templates, and optionally a starting team.
Portfolio Overview	A matrix of principal records (rows) against process stages (columns) whose cells show the status of each stage, for a chosen scope.
Checklist Template	A reusable, versioned list of checklist items kept in the Checklist Library. It stands on its own and may be associated with zero, one or several gates of the Gate Library; when associated, it is copied into the gate checklist when the phase opens.
Vertical	An industry sector (for example Automotive, Food & Beverage, Public Education) defined as a first-class, versioned configuration entity that activates its own macro processes, E2E processes, data extensions, compliance mappings, checklists and gate criteria.
Vertical Instance	The content of one vertical — its prefix, macro and E2E processes, data model, compliance, integration, configuration and governance — delivered as configuration data, not as platform code.
SME	A small or medium-sized enterprise, by default 1–250 employees (EU) or up to 500 (US SBA), thresholds configurable per region, segmented into Micro (1–10), Small (10–50) and Medium (50–250).

Term	Definition
Mode (Full / SME)	The rigor profile of a project: Full mode runs the complete process set; SME mode runs a right-sized SME track.
SME Track	One of two to four configurable levels of process rigor for SME mode, each defining its macro and E2E processes, gates, checklist items per gate and typical duration.
Complexity Score	A weighted score of configurable criteria (weights summing to 100 %), including vertical-specific drivers, that recommends the track of a project.
Project Template Catalog	The governed, versioned library of project templates, each classified by scope (universal or one vertical) and mode (Full or SME, with its track).
Creation Mode	The way a project is created: from the catalog, manually, or with AI (an AI-drafted project reviewed and accepted by a person).
Phase	A stage of a project's process chain (for example an E2E process or a lifecycle stage) to which gates and checklists are attached.
Gate Definition	A reusable decision point — name, criteria, approvers, decision options Go / No-Go / Hold / Recycle — kept in the Gate Library, with zero or more associated checklists, and attachable to any phase.
Pack / Bundle	A priced commercial package: an SME pack, a vertical pack, or a bundle of both.
Questionnaire Template	A versioned, reusable questionnaire — sections and questions — tagged by focus, sector and language, from which a questionnaire is loaded.
Focus	The thematic lens of a questionnaire or engagement: All needs, Digital, AI, or a named Custom focus.
Distribution Channel	The way a questionnaire reaches a respondent: Face-to-Face (with offline capture), Email, Hybrid, or a Custom channel.
Stakeholder	A person registered as a respondent for an engagement, with role, function, decision level, preferred channel and consent status.
Function	A business function of the Organization — for example Quality, Health & Safety, Finance, Human Resources, Operations — to which roles, processes and governance items are linked. Functions are a versioned catalog per Organization.
OBS Role	A position defined in an OBS unit (for example Quality Manager of the head office), linked to one or more functions and played by one or more people. An OBS Role describes what a person does in the organization; it is distinct from an access role of RBAC.
Role Assignment	The fact that a person plays an OBS Role, with a start date, an optional end date, an allocation percentage and a holder type (holder, deputy or acting). A person may hold several Role Assignments at once.
Process Design	The set of typed elements that define how the application's work is done: functions, end-to-end processes and their phases, macro processes, tasks, steps, gates, checklists, step forms, templates and the AI Use Cases linked to them.
Process Element	Any one item of the Process Design, held as data, with a stable identifier and a version history.
Design Release	A named, immutable snapshot of the Process Design, or of one function or process within it, that can be published, compared with another release and restored.
Current Version	The version of a versioned entity that the application uses for new work. Every earlier version stays viewable and can be restored, which creates a new Current Version with its content.
Prompt Specification	The structured prompt of an AI Use Case: Role (persona), Context, Task (instruction), Inputs, Knowledge sources, Constraints, Examples, Output format, Tone and language, Quality criteria, Human checkpoint and Model parameters, each held in its own field and versioned.
Step Form Kind	The type of input form a step uses — for example record table, assessment matrix, decision,

Term	Definition
	objectives, plan, monitoring, review, assignment, document — which decides its fields, its checks and the records its completion creates.
Readiness Checklist	An optional list, shown before a process starts, of what should be in place: inputs, predecessor processes, owner and RACSI, templates, KPIs. Items the system can verify are checked automatically; the list never blocks the start.
Documented Information Template	A versioned template for a controlled document or record (policy, procedure, register, report), made of typed sections — text, table, approval block — each filled from a named data source of the application.
Global Search	One search entry, in the navigation menu, that finds every record, process element, governance item, document and Help topic the user may open.
Document Profile	The minimum content a generated document of a given type must carry — identification block, purpose and scope, method or rating scales, data sections, analysis, actions, revision history, approval — and the checks applied before it is published.
Data Source Binding	The link between a section of a template or document and a named data source of the application (a register, a list of records, the values of a step), with its filters and sort order.
Consistency Check	An automatic verification, run on a generated document and across the documents of a project, that every value, identifier and reference agrees with the application's records and with the other documents.
Section Override	Text or a value typed in a document section that replaces, for that document only, what the data source provides; it is marked as such and kept across regenerations.
Edit-through	Changing a row of a data-bound table from within a document, the change being written to the source record (with its rights, checks and audit) rather than to the document.
Controlled List	A list of proposed values for a field (for example categories, types, parties), managed as configuration data, always offering a Custom value typed by the user.
Choice Library	The values of a Controlled List together with the values users have already typed in the Organization, offered again the next time the field is filled.
Entry Origin	How a record was entered: typed by the user (Manual), taken from a library (Library) or suggested by the AI and accepted by a person (AI).
Record Link	A many-to-many link between two records (for example a need and the interested parties it concerns), each end chosen from the records already identified or typed as a new value.
Register Propagation	The creation or update of register entries from the rows of a completed step, each row keeping the identifier of the entry it produced.
Row Editor	A large window that edits one row of a record table field by field, with scrolling, previous and next row, and a return to the table.
Engine Transparency	The display, with every AI answer, of the engine that produced it (live model or built-in engine) and, when the live model was not used, the reason.

1.5 References

- [journi](#) — Software Requirements Specification, Version 2.0, September 2026.
- [NCP Solver](#) — DynamicMS Suite Software Requirements Specification, Version 2.0, September 2026.
- [Dynamic Apps Suite](#) — Brand & Graphical Chart Guide, Version 1.0, August 2026.
- [CortexPLM](#) — Reference implementation of the Dynamic Apps platform layer, build of September 2026, with its SRS Gap Analysis, September 2026.
- [Application-Agnostic Requirements for Verticals and SME](#) — Vertical and SME layers of any PLM or PSLM platform (29 verticals, SME framework, Vertical × SME intersection, non-functional requirements, traceability and requirements governance), 2026.

- CortexSkills Specs Pack — Master Change Table CR1: Addition of Scope of Work Definition & Questionnaire Management, September 2026.
- DynamicMS — Reference implementation of an integrated management system Dynamic App (QMS and QHSE), build of September 2026, with the SME QMS Scenario 1 feedback on its User Guide 2, September 2026.
- DynamicMS — Feedback rounds 2 and 3 on the September 2026 build (IMS document content, SRS 1.5 features, user experience and Scenario 1 of User Guide 2), September 2026.

1.6 Document Overview

Section 2 describes the Dynamic Apps Suite at a high level and states the design principles every Dynamic App is expected to follow. Section 3 defines the common platform architecture. Section 4 enumerates the common functional requirements, grouped by shared capability and individually identified (FR-DA-xxx) for traceability. Section 5 defines common non-functional requirements (NFR-DA-xxx). Section 6 summarizes the common data-model entities. Section 7 defines common external interface conventions. Appendix A records where Jurni and NCP Solver currently diverge from this standard, so the gap is explicit rather than silently absorbed.

Revision 1.1 adds Sections 3.11 to 3.13 and 4.19 to 4.21, adds requirements to existing sections under new, never-reused IDs, and adds Appendix B, the change log of this revision. No 1.0 requirement was removed or renumbered.

Revision 1.2 adds Sections 4.22 to 4.24, new requirements in Sections 4.1, 4.4, 4.7, 4.12, 4.19, 4.21 and 5.5, and Appendix C, the change log of revision 1.2. As in 1.1, nothing was removed or renumbered.

Revision 1.3 adds Section 3.14, Sections 4.25 to 4.35, new requirements in Sections 4.13, 5.1 to 5.6, 5.8 and 5.10, and Appendices D (change log) and E (traceability to the Verticals and SME source). It renames Section 4.23 Checklist Library and rewords FR-DA-CHK-01 to 05 so that a checklist no longer has to belong to a gate; Section 4.35, the Gate Library, holds the gates and their associated checklists. Nothing was removed or renumbered.

Revision 1.4 adds Section 4.36 Questionnaire & Survey Management and Section 5.12 Questionnaire Data, new requirements in Sections 4.13 and 4.17, five entities in Section 6, 4 terms in Section 1.4, and Appendix F, the change log of revision 1.4. Nothing was removed or renumbered.

Revision 1.5 adds Sections 3.15 to 3.18 and 4.37 to 4.43, new requirements in Sections 4.4, 4.6, 4.16, 4.19, 4.24, 5.1, 5.2, 5.3, 5.5 and 5.8, 12 terms in Section 1.4, 16 entities in Section 6, and Appendix G, the change log of revision 1.5. Nothing was removed or renumbered. NFR-DA-SEC-07 still governs a personal live-model connection; NFR-DA-SEC-17 adds the rules for an Organization-level connection stored encrypted on the server (FR-DA-AI-16).

Revision 1.6 adds Sections 4.44 to 4.48, new requirements in Sections 4.4, 4.19, 4.40, 5.1, 5.2, 5.3, 5.5 and 5.8, 12 terms in Section 1.4, 12 entities in Section 6, rows in Appendix A and Appendix H, the change log of revision 1.6. Nothing was removed or renumbered; FR-DA-DOC-01 to 11 (Section 4.43) remain the base that Sections 4.44 and 4.45 detail.

2. Overall Description

2.1 Product Perspective

A Dynamic App is a self-contained, installable web application — a browser-based single-page frontend paired with a REST API backend — that can run standalone or, per NFR-DA-MAINT-01, expose its composable modules (Action/Task Register, Alert & Notification Engine, RBAC layer, KPI/Reporting engine, Audit Trail Service) for reuse by sibling Dynamic Apps behind a shared API contract. Every Dynamic App is positioned as data- and AI-led: its deliverables are evidence-based records, structured workflows, and governed AI-assisted suggestions, never autonomous decisions.

2.2 Design Principles

- Explainable AI only. Every AI capability is tiered Assistive or Augmented; a third, Autonomous tier — an AI that takes an irreversible action without a human decision in the loop — is out of scope for every Dynamic App.
- Tenant isolation by construction. Every data-bearing record is scoped to an Organization; no query, export, or AI retrieval may return another tenant's data.
- Commercial gating as a second, independent axis. Solution Pack and Add-On/Compliance entitlement gate feature and quota access alongside, never instead of, RBAC.
- Governance is structured, not narrative. Business rules, controls, risks, and accountability are captured as typed, queryable records — tagged back to a process or stage — rather than left in prose documents.
- One shared visual and interaction language. Every Dynamic App, and every document it generates, conforms to the suite's common visual identity (color palette, typography, iconography) across every module, output format, and supported language.
- Multilingual and RTL-first. English, French, and Arabic (full right-to-left layout) are first-class from the outset, not a later localization pass.
- One navigation shell, personal to each user. Every Dynamic App shares one navigation shell; each user may pin the menu or let it slide away, dock it on any screen edge, and mark favorites, and these choices follow the user across sessions and devices.
- Compare within the family, never outside it. Benchmarking compares an Organization with itself and with its own Group, from aggregates only; it never exposes another tenant's records and never reaches beyond the Group.
- Upgrade and recover in place. A new release upgrades an existing installation at startup without data loss, and the data is backed up on a schedule the operator does not have to remember.
- One process design, versioned. Every element of the process design is data with a full history: it can be created, changed, compared and restored to any earlier version, and a running record keeps the version it started with.
- Accountability through roles, not names. Work is assigned to OBS roles, linked to functions and resolved to the people who play them, so a change of person never breaks an assignment.
- Explicit prompts. Every AI Use Case carries its full prompt as separate, reviewable, versioned fields, populated for the exact task or step it serves.
- One truth, many documents. A generated document never holds data of its own that contradicts the application: every value comes from a record, is traceable to it, and is the same in every screen and every document that shows it.

- Usable before complete. A field names what it expects, offers a list with a Custom value, gives room to type, and explains what is computed; the user never meets a generic "Item" or a box too small to read.

2.3 Product Functions (Summary)

- Register, resolve/manage, and evidence a structured record type (an NCP Sheet, a Change Management Project) through a defined multi-stage lifecycle.
- Provide one or more rule-based AI agents grounded in a retrieval corpus, plus a chat-style AI Assistant offering RBAC-scoped data queries and application-help queries.
- Maintain a governed AI Use Case Library — a seeded catalog plus organization-authored custom entries, versioned, with Organization-level and Project-level activation.
- Maintain per-process/per-stage governance: Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, and a RACSI accountability matrix.
- Model process workflows as real BPMN 2.0 diagrams, importable and exportable as standard XML.
- Provide Configuration Management: Solution Packs, priced Add-Ons, Compliance & Security Standards, deployment options, and support tiers, each enforced wherever the platform gates a feature.
- Provide full role-based access control with a runtime-editable Permission Matrix, and an organizational hierarchy (Group/Organization/Project) with an OBS for people/department scoping.
- Compute standard KPIs plus Custom KPIs, and standard Reports exportable to PDF, Excel, and Word; support CSV export on high-traffic tables.
- Operate multi-tenant with data isolation, and multi-language with full Arabic RTL support.
- Provide a searchable, multi-language in-app Help guide covering every module.
- Provide a personalizable navigation shell: menu pinned or sliding, docked left, right, top, or bottom, with favorites, persisted per user.
- Provide internal benchmarking of standard metrics between segments of one Organization and between Organizations of the same Group.
- Protect and upgrade data in place: on-demand and daily backups with retention, and additive migrations at startup.
- Activate verticals and an SME mode by configuration, score project complexity to choose the right track, and start projects from a governed template catalog, by hand, or with AI, with gates and checklists attached to any phase.
- Design questionnaires from a versioned template library, by hand or with AI under human validation, distribute them across channels with reminders, and collect, validate and consolidate consented responses into a tenant-scoped dataset.
- Find any record, process element, document or Help topic the user may open from one search in the navigation menu.
- Manage the process design — functions, processes, phases, tasks, steps, gates, checklists, forms and templates — with full history, comparison and restore of any version.
- Model the organization as units, roles and people, with roles linked to functions and people playing one or more roles.
- Generate controlled documents and records from versioned templates filled with the application's own data, downloadable in Word, PDF and Excel.
- Generate detailed, consistent documents from the application's data, checked before publication and editable at every level — document, version, section and data-bound row — with full history.

- Capture records with qualified fields, controlled lists with Custom, large row editing and many-to-many links, entered manually, from a library or with the AI, and propagate them to registers and documents.

2.4 User Classes and Characteristics

Every Dynamic App defines its own domain-specific role names, but each role maps onto one of the following six baseline classes, which the common Permission Matrix (Section 4.2) is built around:

Baseline Class	Characteristics	Example Mapping
Platform / Organization Administrator	Full configuration authority: Configuration Management, RBAC, hierarchy, AI Use Case governance, Compliance Standards.	Administrator (NCP Solver); Super Admin, Group Admin, Organization Admin (journi)
Process Owner / Manager	Owns the governed process end-to-end: validates records, configures thresholds, owns Controls/Business Rules, full governance CRUD.	Quality Manager, CI Pilot (NCP Solver); Change Manager (journi)
Team Contributor	Creates/edits records and actions within an assigned scope; completes assigned workflow stages.	NCP Team Member (NCP Solver); Practitioner (journi)
Action Owner / Evaluator	Executes or verifies a specific assigned action; owner and evaluator of the same action must always differ.	Action Owner (AR) / Evaluator (AE) (NCP Solver); comparable action-level roles (journi)
Scoped Viewer / Sponsor	Department- or team-scoped visibility into records, dashboards, and governance items relevant to that scope.	Department Head (NCP Solver); Sponsor, People Manager (journi)
Reporter / Read-Only / Auditor	Either submission-only creation with no post-submission edit rights, or full read-only access including the audit trail.	Reporter, Read-Only/Auditor (NCP Solver); Employee, Executive (journi)

2.5 Operating Environment

- Server: a standard Node.js runtime (Node.js 18+ minimum; Node.js 22.13+ recommended — the first long-term-support line whose built-in SQLite module works without an experimental flag — so the platform can use the runtime's own SQLite module rather than a native-compiled dependency). The required version shall be declared in the package manifest.
- Client: any evergreen desktop, tablet, or mobile web browser, with a responsive layout adapting to each; no legacy-browser support is in scope.
- No mandatory external service dependency: a Dynamic App must remain fully functional with no live third-party call, with the sole opt-in exception of a user-configured Real LLM Provider Connection (Section 4.3).
- Report-export and persistence libraries must be pure JavaScript / native-runtime, so installation succeeds without a native C/C++ build toolchain, a required container orchestrator, or an external database server.

2.6 Design and Implementation Constraints

- Every data access must be scoped by tenant (organization_id); no query may return another tenant's data.
- All AI agent behavior must be explainable and logged; no black-box autonomous decision-making.

- The UI, and every generated document, must conform to the suite's common visual identity across every module, output format, and supported language.
- An action's Owner and its Evaluator, wherever the platform models a two-party verification step, must always be different users, enforced server-side.
- Solution Pack entitlement is a second, independent gating axis alongside RBAC, and every Pack quota must be checked server-side at the point of creation/activation.
- Every user-facing string must exist in a single translation dictionary for every supported language; no hardcoded UI string is acceptable outside it.
- Benchmark data leaving an Organization, even towards its own Group, must be aggregated; no record, name, or identifier of another tenant may be returned.

2.7 Assumptions and Dependencies

- Organizations are responsible for their own user provisioning within the tenant they administer.
- Rule-based AI agent implementations are a functional, explainable stand-in that a future LLM/vector-database-backed implementation can replace behind the same internal interface, without changing calling code.
- Compliance & Security Standard modules provide GRC scaffolding only; an Organization remains responsible for its own compliance program and any external certification.
- Demo/seed data is fictional and is not a substitute for real organizational data before go-live.
- Off-site copies of backups, and the hosting platform's own disaster-recovery arrangements, remain the responsibility of the operator of each deployment.

3. Common Platform Architecture

3.1 Tenancy & Hierarchy

Every Dynamic App is built on the same three-to-four-level hierarchy: an optional Group above the Organization (tenant) level, the Organization itself as the hard data-isolation boundary, and one or more Projects below it. An Organizational Breakdown Structure (OBS) — a site/department/service/team tree, or an equivalent per-project resourcing roster — supplies the shared 'who/where' backbone that governance modules (Business Rules, Controls, Risks & Opportunities, RACSI, BPMN diagrams) point back to.

Group membership is also the only boundary across which Organizations may compare performance (Section 3.12). Every hierarchy screen shows whether an Organization belongs to a Group (Yes/No) and which one; an Organization with no Group is shown as Independent.

3.2 Identity & Access Control

A Dynamic App recognizes a fixed catalog of standard roles (nine in the reference implementations) and supports assigning one or more roles per user. Every write/manage capability check reads from a runtime-editable Permission Matrix — a role × capability grid, seeded with sensible defaults, editable in place by a Platform/Organization Administrator, and effective immediately, platform-wide, without a code change or restart. The matrix is a fail-safe override on top of an underlying default model, never a bypass of it.

3.3 Configuration Management & Commercial Gating

A single Configuration Management surface records, per Organization, exactly one active Solution Pack (defining a stock feature/module list, AI tier, and quota ceilings), any number of independently priced Add-On toggles, and any number of independently priced Compliance & Security Standard toggles. Every Pack-gated route and every quota check must be enforced at the server/API boundary, independently of and in addition to RBAC — client-side navigation gating is a UX convenience only, never the security boundary.

3.4 AI Governance Architecture

A governed AI Use Case Library is the single designated integration point for any live LLM call. Every use case is versioned, tiered Assistive or Augmented, activatable per Organization and overridable per Project (Inherit/On/Off), and logged to an append-only AI Usage Log capturing each suggestion's outcome (Accepted/Edited/Rejected). A Real LLM Provider Connection, where offered, is configured and stored client-side only, is forwarded once per generation request, and is never persisted server-side, logged, or included in any export. Every AI Use Case computes a deterministic, explainable result and falls back to it automatically if a live call fails, so a misconfigured or unreachable connection never blocks a suggestion.

3.5 Retrieval-Augmented Generation (RAG) Architecture

AI-assisted suggestions, an AI Assistant's application-help mode, and any 'ask the application' chatbot are grounded by retrieval over one or more corpora — the tenant's own historical records, a Standards/Knowledge Base, the tenant's own Return on Experience (REX) entries (Section 3.10),

and/or a static feature/FAQ corpus — using a TF-IDF/cosine-similarity search or an equivalent lightweight mechanism that runs fully offline, with no network call or API key, so it is reproducible and independently verifiable. RBAC visibility rules applied to retrieval must be the same rule the rest of the application enforces, never a second, hand-maintained copy liable to drift out of sync.

3.6 Internationalization Architecture

A single, versioned string dictionary keyed by string id holds English, French, and Arabic values for every user-facing string. Language precedence resolves, on every sign-in and every Organization switch, in this order: the signed-in user's own explicit preference, then the Organization's configured default language, then English as the platform fallback. Arabic renders full right-to-left — navigation, tables, and charts mirror correctly, not merely the text direction.

3.7 Justification Governance Architecture

Every score or state change to a governed record follows a stage-then-justify pattern: the user stages the new value, writes a justification note, then saves — at which point the new value and its justification are appended together, in one atomic update, to that record's audit/change log. A platform-wide toggle determines whether the justification note is a hard requirement before Save enables, or merely offered and recorded when given.

3.8 Version Management Architecture

Any governed entity type may opt into the platform's generic Version Management service rather than implementing its own ad-hoc history mechanism: on every edit, the service duplicates the entity's current state into an immutable, timestamped version record, advances the entity's Current pointer to the new version, and never rewrites or deletes a prior version. Reverting to a historical version duplicates that version's content forward into a brand-new Current version — the same operation already defined for the AI Use Case Library (Section 4.4) and Template Libraries (Section 4.13) — so the same mechanism backs any entity type an application registers with it, rather than being re-implemented per entity.

3.9 Multi-Channel Communication Architecture

The in-app Notification Center (Section 4.9) is one of several delivery channels a Communication Dispatch service can route a message through — in-app, email, SMS, push, and webhook — behind one shared abstraction, so a new channel can be added without changing the code that raises the underlying alert or message. Every message is composed once, in the recipient's own display language (Section 3.6), and dispatched to every channel the recipient has enabled for that message category; a channel failure is logged and retried on its own schedule and never blocks delivery on another channel.

3.10 Return on Experience (REX) Architecture

Any closable entity — at minimum the application's principal record (a Project, an NCPFiche, a Change Management Project) and an individual Action — can carry a Return on Experience (REX) capture prompted at closure/completion, structured rather than left in free-form prose, and, once saved, indexed into the same retrieval corpus the Standards/Knowledge Base uses (Section 3.5) so a future AI suggestion or AI Assistant answer can be grounded in a prior REX entry exactly as it would be grounded in a Standard or the tenant's own historical records. A REX entry is itself a versioned entity

under the generic Version Management service (Section 3.8), so an edited REX entry keeps its full history like any other governed record.

3.11 Navigation Shell & Personalization Architecture

Every authenticated screen renders inside one shared navigation shell: a primary menu, a header bar carrying the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point and user menu, and the content area. The menu is generated from a single navigation model in which each item declares its module group, route, icon, and the view permission and Solution Pack entitlement it requires — the same codes the server enforces on the route — so an item a user cannot open is never shown, and a new module appears in navigation by being registered once.

Each user chooses where the menu is docked (left, right, top, or bottom) and whether it is pinned open or slides away. Unpinned, the menu collapses to an edge handle and slides over the content when pointed at, focused, or toggled. These choices, the user's favorite items, and collapsed menu groups are stored server-side as user preferences, so they follow the user to any browser or device. Left and right are logical sides: under Arabic RTL they mirror. Below a mobile breakpoint the shell switches to a drawer, whatever the stored dock position.

3.12 Internal Benchmarking Architecture

Internal benchmarking computes a catalog of comparable metrics — each defined once with its unit, its direction of good (higher or lower is better), and its computation from the platform's own records, including the audit trail where history matters — and aggregates them by segment. Two scopes exist. Within an Organization, segments are dimensions of the Organization's own data (for example record or project type, process track, or owning department), and authorized users keep full drill-down to the underlying records. Across a Group, each participating Organization contributes only per-metric aggregates; no record, name, or identifier leaves its tenant.

Every segment or Organization below the minimum sample size is flagged as not comparable and excluded from ranks and medians. An Organization Administrator can withdraw the Organization from Group sharing at any time, after which its figures are no longer shown to the rest of the Group. Comparison with organizations outside the Group, or with external or third-party data, is out of scope.

3.13 Platform Operations Architecture

Operational robustness belongs to the platform layer, not to each deployment. On every start the server applies additive migrations — creating missing tables and columns and adding new permission codes with their default role grants — without dropping data or overwriting an administrator's edits, so an existing installation upgrades in place. Startup guards report an unsupported runtime, a port already in use, or an empty database, each as a plain-language message naming the command that fixes it, rather than a stack trace.

The persistence layer takes consistent online backups on demand and on a daily schedule, keeps them for a configurable retention period, and lists them to the platform administrator. A short-lived, per-user read cache, invalidated by any write in the same Organization, keeps read latency within target under concurrent load without ever serving one user's or one tenant's data to another.

3.14 Vertical & SME Layer Architecture

Requirements are organized in five layers, each built on the one below: Layer 1 Platform (tenancy, security, operations), Layer 2 Core (the common requirements of this standard), Layer 3 Vertical (sector content), Layer 4 SME Track (right-sized rigor for small and medium enterprises) and Layer 5 Packaging & Pricing. A vertical and an SME track are configuration data interpreted by the platform: adding, versioning, activating, rolling back or retiring them never requires a change to platform code or downtime.

Activation is additive and ordered. An Organization activates zero or more verticals; each activates its macro processes (three to five) and E2E processes (two to four). When an SME selects a vertical, the vertical processes are activated first and the SME track processes second. A project then takes a mode (Full or SME), a track recommended by its complexity score, and a template from the catalog; the template brings the phases, the Gate Library supplies the gates of each phase, and each gate brings the checklists associated with it from the Checklist Library; a checklist can also be attached to a phase without any gate.

3.15 Process Design & Versioning Architecture

The process design is a hierarchy of typed Process Elements held as data: functions, end-to-end processes and their phases, macro processes, tasks and steps, the gates and checklists attached to phases, the form of each step, the templates (project, checklist, document, questionnaire) and the AI Use Cases linked to tasks and steps. Every element is registered with the generic Version Management service (Section 3.8): an edit creates a new version, no version is ever rewritten, and restoring any earlier version duplicates it forward as the new Current Version.

Elements can be grouped into a Design Release, a named, immutable snapshot of the whole design or of one function or process, which is published, compared and restored as one unit. A running record is bound to the element versions that were current when it started, or when its phase opened; a later version applies to new records, and to running ones only when an authorized user migrates them with a justification. Completed work is never altered by a change of design.

3.16 OBS Role Model Architecture

The OBS has three layers: units (sites, departments, services, teams, project teams), roles defined within units, and people. A role is linked to one or more functions and carries its responsibilities; a person plays one or more roles through dated Role Assignments. Ownership, RACSI letters, approvers, gate approvers and notification targets are assigned to roles and resolved, at the time of use, to the people who currently play them, so a change of holder moves open work without re-assigning it item by item.

OBS Roles describe what people do in the organization. They are distinct from the access roles of RBAC (Section 3.2): a role may propose default access roles to its holders, but access is always decided by the Permission Matrix.

3.17 AI Prompt Specification Architecture

Every AI Use Case is linked to the exact tasks or steps it supports and carries a Prompt Specification made of separate fields. At run time the platform assembles the prompt from these fields in a fixed order, resolves its input variables from the task or step and the record, adds the retrieved references (Section 3.5), and sends it to the configured live model or to the built-in engine. The specification is

versioned as a whole and field by field, under Section 3.8, and the version used is written to the AI Usage Log with every suggestion.

3.18 Global Search Architecture

One search service indexes, per tenant, the principal records, the process design, governance items, registers, documents, attachments metadata and Help topics, in every supported language. It applies the same RBAC and Solution Pack rules as the screens, through the same code (Section 3.5), so a result, a count or a snippet never reveals something the user could not open. The navigation menu carries its entry point in every dock position.

4. Common Functional Requirements

Each requirement below is common to journi and NCP Solver, generalized to platform vocabulary and stated in “the system shall” form for verifiability. Where the two source SRS documents diverge on a requirement's implementation depth, the stronger (more complete) formulation is taken as the standard; Appendix A records where a given application's current build falls short of it.

Requirements added in revision 1.1 are generalized from the CortexPLM reference build; they carry new IDs following the last 1.0 ID of their section, are listed in Appendix B, and their conformance status is recorded in Appendix A.

Requirements added in revision 1.3 in Sections 4.25 to 4.32 restate the Application-Agnostic Requirements for Verticals and SME source in platform vocabulary; duplicates in the source (for example the vertical approval workflow and audit trail stated twice) are merged, and Appendix E gives the mapping. Sections 4.33 to 4.35 come from the Dynamic Apps product direction.

Requirements added in revision 1.5 are generalized from the DynamicMS reference build and from its user feedback; they carry new IDs, never reuse an earlier one, and are listed in Appendix G.

Requirements added in revision 1.6 come from the second and third rounds of DynamicMS user feedback; they carry new IDs, never reuse an earlier one, and are listed in Appendix H.

4.1 Tenant & Organizational Hierarchy

- **FR-DA-TEN-01:** The system shall support full Create, Read, Update, and Delete on Group, Organization, and Project records, gated by a hierarchy-management capability.
- **FR-DA-TEN-02:** The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully independent.
- **FR-DA-TEN-03:** Deleting a Group shall not delete its member Organizations; deleting an Organization shall cascade-delete its Projects and directly-scoped user accounts and AI activation/override state.
- **FR-DA-TEN-04:** The system shall support full Create, Read, Update, and Delete on Organizational Breakdown Structure (OBS) entries, each typed or free-text, optionally parented to another OBS entry, forming a per-Organization or per-Project tree.
- **FR-DA-TEN-05:** The system shall prevent deleting an OBS node from breaking referential integrity, either by cascading or by re-parenting/reassigning dependent records.
- **FR-DA-TEN-06:** The system shall display, for every OBS node, a live count of linked item types, so accountability and activity concentration are visible at a glance.
- **FR-DA-TEN-07:** The system shall enforce the active Solution Pack's maximum-Projects and maximum-OBS-node quotas at creation time, server-side, rejecting an over-quota request with HTTP 409.
- **FR-DA-TEN-08:** The system shall ensure that no Organization can enumerate, query, or infer the existence of another Organization through any API response, error message, or shared identifier sequence.
- **FR-DA-TEN-09:** The system shall display, on every Organization list and detail screen, whether the Organization belongs to a Group (Yes/No) and, if so, which Group; an Organization with no Group shall be labeled Independent.
- **FR-DA-TEN-10:** The system shall reject, server-side, an OBS parent assignment that crosses trees — an Organization-level node under a Project-level node, or a node under another Project's tree — and an assignment that makes a node its own ancestor.

- **FR-DA-TEN-11:** The system shall identify records by values that reveal nothing about other tenants' activity — opaque identifiers (e.g. UUIDs) or per-tenant sequences — and shall answer a request for another tenant's record exactly as for a record that does not exist (HTTP 404), in support of FR-DA-TEN-08.
- **FR-DA-TEN-12:** The system shall present a tenancy tree, Group (Yes/No) > Organization > principal records, and the same content as a table (Group Yes/No, Group, Organization, sector, records), limited to the Organizations the viewer may see.
- **FR-DA-TEN-13:** The system shall let an authorized user choose the Organization a new principal record belongs to before creating it, showing whether that Organization belongs to a Group; a platform administrator may choose any Organization, other users only their own.
- **FR-DA-TEN-14:** The system shall provision every new Organization on creation with the OBS skeleton, the standard governance catalog, the AI Use Case Library (FR-DA-AI-02), the checklist and record templates, and, when requested, a starting team of one account per standard role under the Organization's e-mail domain with an initial password; a domain already used by another Organization shall be refused.
- **FR-DA-TEN-15:** The system shall let a dedicated permission grant read-only visibility of the principal records of the other Organizations of the viewer's own Group; records of Organizations outside the Group shall never be visible, and records of another Organization shall never be editable from the viewer's Organization.
- **FR-DA-TEN-16:** The system shall let an OBS node list the people placed in it, each with a standard role and the role played in that node (for example "Engineering lead" in a project team), and shall let an administrator add and remove them.

4.2 Identity & Role-Based Access Control

- **FR-DA-RBAC-01:** The system shall support full Create, Read, Update, and Delete on user accounts, including name, email/username, one or more assigned roles, and language preference.
- **FR-DA-RBAC-02:** The system shall provide a Permission Matrix screen displaying every permission/capability code against every role, editable by users holding a dedicated permission-management capability, with edits taking effect immediately and platform-wide without a code change.
- **FR-DA-RBAC-03:** The system shall enforce every API endpoint, server-side, against the authenticated user's resolved permission set (the union across all assigned roles); a request lacking the required permission shall be rejected with HTTP 403.
- **FR-DA-RBAC-04:** The system shall enforce that the Owner and the Evaluator of any two-party verification step are never the same user, both in the UI and at the server.
- **FR-DA-RBAC-05:** The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assignment purposes, independent of the full user-management permission.
- **FR-DA-RBAC-06:** The system shall issue a signed, time-boxed authentication token on successful login, required on every subsequent API call, and shall never expose a stored password hash or credential via any API response.
- **FR-DA-RBAC-07:** The system shall expose a single platform-wide justification-required toggle (Section 3.7), defaulting to on, gated by a hierarchy/governance-management capability.
- **FR-DA-RBAC-08:** The system shall, when a user opens a screen or deep link that their roles or the Organization's configuration do not allow, show an in-app access notice explaining why and whom to ask, never a blank page or a raw error.
- **FR-DA-RBAC-09:** The system shall provide a dedicated permission for viewing the content of individual evaluation or verification verdicts; other users, except the parties to that verification, shall see that an evaluation exists but not its content.

- **FR-DA-RBAC-10:** The system shall, at startup, add any permission code introduced by a new release to the catalog together with its default role grants, without overwriting grants an administrator has changed (Section 3.13).

4.3 Configuration Management (Solution Packs, Add-Ons, Compliance Standards)

- **FR-DA-CFG-01:** The system shall record, per Organization, exactly one Solution Pack as the single source of truth for that Organization's feature and quota entitlements, defined centrally in one configuration module and consumed by every enforcing route, so entitlements are never duplicated or able to drift between routes.
- **FR-DA-CFG-02:** The system shall define, for each Solution Pack, at minimum: which modules/features are enabled, the AI tier available (Assistive-only vs. Assistive+Augmented), and numeric ceilings on Projects, OBS nodes, and AI Use Cases.
- **FR-DA-CFG-03:** The system shall record a set of priced Add-Ons, each an independent toggle that grants its associated feature to a Pack that does not already include it standard.
- **FR-DA-CFG-04:** The system shall record a set of priced Compliance & Security Standards modules (at minimum GDPR, ISO/IEC 27001, and SOC 2), each an independent toggle available regardless of Solution Pack.
- **FR-DA-CFG-05:** The system shall, on activating a Compliance & Security Standard, idempotently seed a starting Controls scaffold tagged to that framework, without duplicating it on repeated activation, and shall not delete any Control tagged to a Standard that is later deactivated.
- **FR-DA-CFG-06:** The system shall enforce every Pack-gated feature and every Add-On/Compliance-gated feature server-side via dedicated middleware on the corresponding API route, independently of and in addition to RBAC permission checks; a request for a feature the active configuration does not include shall be rejected with HTTP 403, naming the missing feature.
- **FR-DA-CFG-07:** The system shall expose the Organization's effective configuration and the static Pack/Add-On/Compliance catalog to every authenticated user regardless of role, so client-side navigation can correctly show or hide gated menu items; this read is a UX convenience, not the enforcement boundary.
- **FR-DA-CFG-08:** The system shall expose current usage against every quota dimension, computed live from current data, and shall visually flag a quota that meets or exceeds its allotment.
- **FR-DA-CFG-09:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **FR-DA-CFG-10:** The system shall allow an authorized Organization administrator to register, configure, enable, and disable a connection to an external application through the External Integration Registry (Section 3.3), independently of any other registered integration.
- **FR-DA-CFG-11:** The system shall store every external integration's credentials using the same client-side-only or encrypted-server-side pattern already required for the Real LLM Provider Connection (NFR-DA-SEC-07), never in plaintext, and never logged.
- **FR-DA-CFG-12:** The system shall verify an external integration's connection health on demand and shall record the outcome (success, authentication failure, unreachable) with a timestamp, without exposing provider-side error detail that could aid an unrelated attacker.
- **FR-DA-CFG-13:** The system shall allow an authorized administrator to map fields between an external application's records and the platform's own entities, and shall log every inbound or outbound integration transaction (record, timestamp, direction, and result) for audit (Section 4.11).
- **FR-DA-CFG-14:** The system shall accept inbound events from an external application only through the Multi-Channel Communication webhook channel (Section 4.17) or an equivalent authenticated endpoint,

never an unauthenticated route, and shall scope every inbound event to the Organization that registered the originating integration.

4.4 AI Use Case Library & Governance

- **FR-DA-AI-01:** The system shall support full Create, Read, Update, Delete, and version-revert on an AI Use Case catalog, each entry capturing an id, name, tier (Assistive/Augmented), owning module, trigger, expected output, a human-checkpoint description, and versioned prompt-template content, gated by a dedicated AI-use-case-management capability.
- **FR-DA-AI-02:** The system shall seed every Organization with the same library of AI Use Cases on creation, and shall allow an Organization to additionally author custom AI Use Cases up to its active Solution Pack's custom-use-case quota.
- **FR-DA-AI-03:** The system shall create Version 1 of an AI Use Case automatically on creation, mark it Default/Current, and shall retain the full version history indefinitely; reverting to a historical version shall duplicate its content into a brand-new Current version, never rewrite or delete history.
- **FR-DA-AI-04:** The system shall allow an Organization Administrator to activate or deactivate any AI Use Case for the whole Organization, and shall allow an authorized Project-level user to set a tri-state (Inherit/On/Off) override per Project; effective activation shall resolve the Project override when set, else the Organization's activation state.
- **FR-DA-AI-05:** The system shall block, server-side, any new AI suggestion from a deactivated AI Use Case (by its effective state) immediately on deactivation, while never retroactively hiding or removing content a human already reviewed and approved before deactivation.
- **FR-DA-AI-06:** The system shall visually label every AI-generated output, wherever shown, as AI-generated and requiring human review, alongside its tier and its generation source (deterministic built-in engine or a configured live model).
- **FR-DA-AI-07:** The system shall write an append-only AI Usage Log entry for every AI suggestion's outcome (Accepted, Edited, or Rejected), capturing the use case, Organization, Project, the record it applied to, the acting user, and a timestamp; the log shall expose no edit or delete route.
- **FR-DA-AI-08:** The system shall allow a user to configure a Real LLM Provider Connection (provider, model, endpoint, API key) that is persisted client-side only, is forwarded once per generation request, is never returned in any API response, never written to any log, never cached or persisted server-side beyond that single request, and is never included in any data export.
- **FR-DA-AI-09:** The system shall, on any failure of a live LLM call, fall back automatically to the AI Use Case's deterministic built-in generator, so a misconfigured or unreachable connection never blocks a suggestion from being produced.
- **FR-DA-AI-10:** The system shall ensure an AI Use Case's activation, tier, or generation never grants a user visibility into data beyond what their RBAC role and permissions already allow.
- **FR-DA-AI-11:** The system shall, before generating any suggestion, ground it in retrieved reference content (Section 4.5), instruct the underlying model to stay within the application's own vocabulary, and state no number, score, or name not present in the retrieved references or the record's own data; the retrieved references shall be disclosed to the user alongside the output.
- **FR-DA-AI-12:** The system shall show every active AI Use Case, labelled with its tier (Assistive AI or Augmented AI) by a distinct icon and wording, on the processes, macro processes, tasks and steps it is linked to, and shall let an authorized user run it there and accept, modify or reject the suggestion (FR-DA-AI-07).
- **FR-DA-AI-13:** The system shall let an AI Use Case be linked to a step, a process, a task or an end-to-end process, and shall list the tasks of the process design with the AI Use Cases that support them, filterable by tier.

- **FR-DA-AI-14:** The system shall let an authorized user delete a custom AI Use Case, keeping its usage-log entries; seeded catalog entries shall be deactivated instead of deleted.
- **FR-DA-AI-15:** The system shall offer, wherever the live model is configured (personal settings and administration), a provider list and a model list with a "custom model" choice and an optional custom endpoint, and a connection test that reports only the outcome category (works, authentication failure, model not found, rate limited, unreachable).
- **FR-DA-AI-16:** The system shall let an Organization Administrator configure an Organization-level live model — provider (standard providers or a custom endpoint compatible with a common API), model, temperature and maximum answer length — as an alternative to the personal connection of FR-DA-AI-08; its key shall be stored encrypted under NFR-DA-SEC-17, shown masked, and never returned in full.
- **FR-DA-AI-17:** The system shall let an authorized user choose, per AI Use Case, a model of the configured provider other than the Organization default.
- **FR-DA-AI-18:** The system shall let a user view, before and after a run, the exact prompt the AI Use Case sends — assembled from its Prompt Specification and resolved for the current task or step — together with the engine that answered (live model or built-in engine) and any fallback warning.
- **FR-DA-AI-19:** The system shall keep a capability profile per model — the request parameters it accepts (for example sampling parameters), whether it reasons before answering, its maximum output — and send each request with only the parameters the model accepts, reserving enough output length for its reasoning.
- **FR-DA-AI-20:** The system shall offer a connection test for the configured model that reports, in plain language, whether the key was refused, a parameter was rejected, the model declined the request or the model answered, and record the result.

4.5 Standards / Knowledge Base & Retrieval-Augmented Generation

- **FR-DA-KB-01:** The system shall maintain a Standards/Knowledge Base of reference content — regulatory or methodological standards, or per-process/per-stage practitioner guidance — available in every language the platform supports.
- **FR-DA-KB-02:** The system shall index this knowledge base, the tenant's own historical records, and a static application-feature/FAQ corpus using the same retrieval mechanism, so AI Use Case grounding, an AI Assistant's application-help mode, and any chatbot share one retrieval engine rather than several.
- **FR-DA-KB-03:** The system shall scope every retrieval query to the requesting user's own tenant; no cross-tenant record shall appear in a retrieval result or an AI suggestion.
- **FR-DA-KB-04:** The system shall gracefully degrade when a live LLM call is unavailable or fails: the retrieval result (the matched reference content, the computed fact) shall still be returned, rather than the request failing outright.

4.6 Process/Stage Governance (Alerts, Business Rules, Controls, Risks, RACSI, Custom KPIs)

- **FR-DA-GOV-01:** The system shall support full Create, Read, Update, and Delete on Business Rules, each with a condition, an action, a severity, and an owner, gated by a governance-management capability and, where the platform separates GRC from the base module set, by the active Solution Pack's GRC entitlement.
- **FR-DA-GOV-02:** The system shall support full Create, Read, Update, and Delete on Controls, each classified under one of the five COSO Internal Control — Integrated Framework components, with a control type, testing frequency, owner, and effectiveness rating.

- **FR-DA-GOV-03:** The system shall allow a Control to be optionally tagged with the Compliance & Security Standard it was seeded under or otherwise supports, without that tag itself constituting a certification (Section 4.3, FR-DA-CFG-09).
- **FR-DA-GOV-04:** The system shall support full Create, Read, Update, and Delete on Risk/Opportunity items, each with a type, a likelihood (1–5) and impact (1–5) rating, an automatically computed score, an owner, and a status, and shall render a 5×5 likelihood × impact heat-map.
- **FR-DA-GOV-05:** The system shall support full Create, Read, Update, and Delete on Custom KPIs, additive to the platform's built-in KPI set, each with a plain-language formula description, a target, and an owner, available regardless of Solution Pack.
- **FR-DA-GOV-06:** The system shall support a RACSI accountability matrix: full Create, Read, Update, and Delete on RACSI Activities, each optionally linked to a process step or to a specific Business Rule, Control, or Risk/Opportunity record, and on RACSI Assignments, each assigning exactly one of the five RACSI roles to a named person or a role.
- **FR-DA-GOV-07:** The system shall enforce that a RACSI Activity has at most one Accountable assignment at any time, both in the UI and at the data-persistence layer.
- **FR-DA-GOV-08:** The system shall allow a Business Rule, a Control, a Risk/Opportunity, and a RACSI Activity to each be optionally tagged with the owning OBS node and with the process/stage they are most relevant to, and shall surface the items tagged to a given process/stage on that process/stage's own governance view.
- **FR-DA-GOV-09:** The system shall let a RACSI assignment target an OBS Role as well as a named person, and shall resolve a role assignment to the people currently playing that role (Section 3.16) wherever the RACSI is used.
- **FR-DA-GOV-10:** The system shall provide a RACSI editor on every macro process, with one row per task or step and one column per RACSI letter, enforcing exactly one Accountable per row (FR-DA-GOV-07) and defaulting the owner of each step to its Responsible role.

4.7 BPMN 2.0 Process Modeling

- **FR-DA-BPMN-01:** The system shall support full Create, Read, Update, and Delete on BPMN Diagrams, each with a title, description, and a BPMN 2.0-compliant XML payload, rendered/edited with a standards-compliant BPMN 2.0 modeler/viewer component, not a static image.
- **FR-DA-BPMN-02:** The system shall grant read-only pan/zoom viewing of a BPMN Diagram to any user holding view access, and shall grant the full editing palette and a Save action only to users holding edit access whose Organization's active configuration includes full BPMN editing.
- **FR-DA-BPMN-03:** The system shall render the BPMN editor's tool palette in a dedicated panel beside the diagram canvas, structurally separate from the canvas, rather than floating over it.
- **FR-DA-BPMN-04:** The system shall provide Export (download the current diagram as a schema-valid .bpmn/XML file) and Import (load a .bpmn/XML file, replacing the current diagram) actions, so a diagram is portable to and from any other BPMN 2.0-compliant tool.
- **FR-DA-BPMN-05:** The system shall be seeded, per Organization, with at least one BPMN Diagram modeling the application's own core process end-to-end.
- **FR-DA-BPMN-06:** The system shall offer a shapes palette section grouped by category (tools, events, activities, gateways, data, participants, artifacts), usable by click or drag, and shown as a legend of the main shapes when the diagram is view-only.
- **FR-DA-BPMN-07:** The system shall let the user open a diagram in a full-screen workspace with the palette, zoom controls (slider, zoom in and out, fit), a palette that slides away, and a single action or the Escape key to return to the page.

- **FR-DA-BPMN-08:** The system shall state on the diagram screen whether editing a diagram changes the execution of records; where diagrams only document the process, saving one shall not alter running or future records.

4.8 Reports & Data Export

- **FR-DA-REP-01:** The system shall compute the platform's standard KPIs in real time from current data, scoped to the requesting user's tenant.
- **FR-DA-REP-02:** The system shall provide a set of standard reports, each exportable on demand as a downloadable PDF, Excel (.xlsx), and Word (.docx) file, generated server-side and styled to the suite's common visual identity.
- **FR-DA-REP-03:** The system shall name every exported report file with the report key and the export date, and shall serve it with the correct MIME type and a Content-Disposition attachment header.
- **FR-DA-REP-04:** The system shall offer a client-side CSV export on its highest-traffic tables, reflecting exactly the rows and columns currently on screen and respecting the same RBAC and scope filtering as the table itself, written with a UTF-8 byte-order mark so accented and Arabic content opens correctly in common spreadsheet tools.

4.9 Notification Center (Alerts)

- **FR-DA-ALT-01:** The system shall define a shared Alert Catalog naming every alert type once, with a human-readable name and a primary process/stage tag, consumed identically by seed data and by live computation, so alert semantics cannot drift between demo data and live behavior.
- **FR-DA-ALT-02:** The system shall surface a persistent, dismissible per-tenant or per-project alert inbox, with a live unread-alert indicator in the primary navigation.
- **FR-DA-ALT-03:** The system shall avoid raising a duplicate alert of the same type against the same triggering entity within the same period.
- **FR-DA-ALT-04:** The system shall allow a user to mark an individual alert as read, and, where dismissal is supported, shall persist the dismissal separately from the alert's own (recomputed) state.
- **FR-DA-ALT-05:** The system shall allow an authorized user to independently enable or disable each alert type, displaying its human-readable name alongside its machine code in both the alert inbox and any governance-settings configuration list.

4.10 AI Assistant / Chatbot

- **FR-DA-AST-01:** The system shall provide a data-query mode that answers natural-language questions about the requesting user's own organization by matching the question to a fixed set of data-query intents.
- **FR-DA-AST-02:** The system shall associate every data-query intent with the single RBAC permission required to answer it, and shall evaluate that permission after matching the intent but strictly before running any query; when the permission is missing, the system shall return an explicit, human-readable refusal naming the missing permission rather than executing the query.
- **FR-DA-AST-03:** The system shall scope every data-query answer to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-AST-04:** The system shall provide an application-help mode answering “Can it...?” / “How do I...?” questions by ranking a static FAQ/feature corpus with the same retrieval engine used elsewhere in the platform (Section 4.5).
- **FR-DA-AST-05:** The system shall render the AI Assistant both as a dedicated screen and as a floating widget accessible from every authenticated screen, without navigating away from the user's current work.

- **FR-DA-AST-06:** The system shall gate the AI Assistant behind a single view permission and the active Solution Pack's AI Assistant entitlement, independent of and in addition to any module-specific permission a given data-query intent may separately require.

4.11 Audit Trail & Justification Governance

- **FR-DA-AUD-01:** The system shall write an audit or change-log entry for every state-changing operation on a governed field, capturing the actor, timestamp, entity type/id, and the before/after value.
- **FR-DA-AUD-02:** The system shall follow the stage-then-justify pattern (Section 3.7) for every score or state change to a governed record: staging a new value, requiring or offering a justification note per the platform-wide toggle, and appending both atomically to the record's audit/change log in one update.
- **FR-DA-AUD-03:** The system shall make the audit/change log viewable, read-only, to users holding an audit-view permission, and shall never expose an edit or delete route for it.
- **FR-DA-AUD-04:** The system shall ensure audit-log writes never block or fail the primary operation they accompany.
- **FR-DA-AUD-05:** The system shall allow an authorized user to reopen a completed workflow step before the stage or approval it belongs to is decided, only with a justification recorded per Section 3.7; the reopening shall return the step to its active state and be written to the audit log.

4.12 Help & In-App Guidance

- **FR-DA-HLP-01:** The system shall provide an in-app Help screen presenting topic articles covering every module of the application, accessible to every authenticated user independent of role or permission grants.
- **FR-DA-HLP-02:** The system shall render Help content in the user's currently selected display language, matching the article set one-to-one across every supported language.
- **FR-DA-HLP-03:** The system shall provide a free-text search box on the Help screen filtering topics by a case-insensitive match against the topic title or body.
- **FR-DA-HLP-04:** The system shall provide, next to every scored or classified input (for example complexity scores), a decision matrix describing each level, and shall show a legend for every code or colour vocabulary displayed in a list or matrix (for example the Part letter of a process or the status colours of a matrix).

4.13 Template Libraries

- **FR-DA-TPL-01:** The system shall support a library of pre-filled starting points (templates) for its principal record type, each capturing default field values a new record can be seeded from, while leaving every field editable before submission.
- **FR-DA-TPL-02:** The system shall support full Create, Read, Update, and Delete on templates, versioned via the same version-history mechanism as the AI Use Case Library (Section 4.4), gated by a dedicated template-management capability.
- **FR-DA-TPL-03:** The system shall make templates available regardless of Solution Pack, consistent with the always-standard treatment of Custom KPIs and standard Reports.
- **FR-DA-TPL-04:** Deleting a template shall not affect any record already created from it; a template is a starting point copied at use time, not a live-linked reference.
- **FR-DA-TPL-05:** The system shall manage project templates through the Project Template Catalog of Section 4.33; the rules of FR-DA-TPL-01 to 04 apply to it.
- **FR-DA-TPL-06:** The system shall seed a questionnaire template library covering each focus — All needs, Digital, AI and Custom — and each supported sector, following the rules of FR-DA-TPL-01 to 04.

- **FR-DA-TPL-07:** The system shall allow an Organization to author its own questionnaire templates, visible only within its tenant, alongside the seeded library.
- **FR-DA-TPL-08:** The system shall version every questionnaire template and retain its full history through the generic Version Management service (Section 4.16); a questionnaire shall record the template version it was loaded from.
- **FR-DA-TPL-09:** The system shall allow a questionnaire template to be exported as a portable file and re-imported, in the same or another tenant, as a new template version or a new template, with its focus, sector and language tags preserved.

4.14 Multi-Language & Localization & Currency

- **FR-DA-I18N-01:** The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and messages, with no hardcoded UI string outside the shared translation dictionary.
- **FR-DA-I18N-02:** The system shall allow a user to change their display language at any time without reloading or losing unsaved work.
- **FR-DA-I18N-03:** The system shall persist a user's language preference and apply it automatically on subsequent logins, resolved per the precedence order in Section 3.6.
- **FR-DA-I18N-04:** The system shall render the entire application right-to-left, including navigation, forms, tables, and charts, when Arabic is selected.
- **FR-DA-I18N-05:** The system shall allow an Organization to configure a tenant-level default language.
- **FR-DA-I18N-06:** The system shall allow adding a new supported language by configuration — populating the shared translation dictionary (Section 3.6) — alone, without a code change or a redeployment of application logic.
- **FR-DA-I18N-07:** The system shall return every server-generated, user-facing message — validation and error messages, notification text, AI Assistant answers, and export labels — in the requesting user's display language, resolved from the same translation dictionary as the UI; a machine code may accompany the translated text but shall not replace it.
- **FR-DA-I18N-08:** The system shall seed its reference and Knowledge Base content in every supported language and shall let the user filter Knowledge Base articles by language, defaulting to the user's display language.
- **The application should be multi-currency**

4.15 Work Breakdown Structure & Gantt Scheduling

Any grouping of Actions — regardless of the process, stage, or governed record they originate from — can be organized into a Work Breakdown Structure (WBS) and scheduled as a Gantt chart, so planning is available for any action grouping the user assembles, not only a single predefined project type.

- **FR-DA-WBS-01:** The system shall allow any set of Actions the user selects to be grouped into a Work Breakdown Structure (WBS), organized as a tree of WBS nodes, each node referencing one or more Action IDs as its leaf-level work items.
- **FR-DA-WBS-02:** The system shall allow a WBS node or a referenced Action to carry a planned start date, planned end date or duration, percent-complete, and zero or more predecessor dependencies on other WBS nodes or Actions within the same WBS.
- **FR-DA-WBS-03:** The system shall render a WBS as a Gantt chart, plotting every node and its referenced Actions on a shared timeline, drawing dependency lines between predecessor and successor items, and visually distinguishing planned, in-progress, completed, and overdue items.

- **FR-DA-WBS-04:** The system shall recompute a WBS node's own planned dates and percent-complete from its child nodes/Actions when the node itself is not directly scheduled, so a summary bar always reflects the current state of its children without manual reconciliation.
- **FR-DA-WBS-05:** The system shall support full Create, Read, Update, and Delete on WBS structures, and shall allow the same Action to be referenced from more than one WBS without duplicating the underlying Action record.
- **FR-DA-WBS-06:** The system shall export a WBS/Gantt view to at least one portable format (e.g. PDF or an image), consistent with the platform's standard Reports & Data Export conventions (Section 4.8).

4.16 Version Management

- **FR-DA-VER-01:** The system shall provide a single, generic Version Management service (Section 3.8) that any governed entity type can register with, rather than each entity type implementing its own history mechanism.
- **FR-DA-VER-02:** The system shall create an initial version automatically when a versioned entity is created, mark it Current, and create a new version — never overwrite the prior one — on every subsequent edit.
- **FR-DA-VER-03:** The system shall retain every version of a versioned entity indefinitely, with no hard-delete route for a historical version, and shall allow an authorized user to view the full version history and compare any two versions field-by-field.
- **FR-DA-VER-04:** The system shall allow an authorized user to revert a versioned entity to any historical version; reverting shall duplicate that version's content into a brand-new Current version, never rewrite or delete the entity's history, consistent with the pattern already defined for the AI Use Case Library (FR-DA-AI-03) and Template Libraries (FR-DA-TPL-02).
- **FR-DA-VER-05:** The system shall record, on every version, the acting user, a timestamp, and — where the platform-wide justification toggle (Section 3.7) requires it — the justification note that accompanied the change.
- **FR-DA-VER-06:** The system shall provide one shared version-comparison view, available from the history of every versioned entity, that shows two selected versions side by side and highlights the fields that differ, in support of FR-DA-VER-03.
- **FR-DA-VER-07:** The system shall list, on every versioned entity, all its versions with number, label, author, date, change note and justification, and shall let an authorized user open any version read-only and restore it (FR-DA-VER-04) in one action.
- **FR-DA-VER-08:** The system shall version a composite entity — for example a macro process with its tasks and steps, or a gate with its associated checklists — so that restoring one of its versions restores its children, their order and their links as they were in that version.
- **FR-DA-VER-09:** The system shall support named baselines (Design Releases, Section 3.15) grouping chosen versions of several entities, which can be compared with each other and restored as one unit.

4.17 Multi-Channel Communication

- **FR-DA-COMM-01:** The system shall support delivering a notification or alert (Section 4.9) through one or more channels — at minimum in-app, email, and webhook, with SMS and push as optional channels — behind a single shared dispatch interface (Section 3.9).
- **FR-DA-COMM-02:** The system shall allow a user to independently enable or disable each available channel for each notification category, defaulting every category to the in-app channel at minimum.
- **FR-DA-COMM-03:** The system shall compose a message once, per notification event, and render it in the recipient's own display language (Section 3.6) before dispatch to any channel.

- **FR-DA-COMM-04:** The system shall record a delivery-status record per message per channel (queued, sent, delivered, failed), and shall retry a failed delivery on that channel's own schedule without blocking or duplicating delivery on any other channel.
- **FR-DA-COMM-05:** The system shall never send a tenant's data through an external channel provider (email, SMS, push) beyond the minimum content required for that notification, and shall scope every dispatch to the recipient's own tenant.
- **FR-DA-COMM-06:** The system shall provide a dedicated questionnaire dispatch channel abstraction on top of the Communication Dispatch service (Section 3.9), supporting Face-to-Face (printable or offline packs), Email, Hybrid and Custom channels, so that a new questionnaire channel can be added without changing questionnaire logic.
- **FR-DA-COMM-07:** The system shall let an authorized user set a reminder cadence per channel for each questionnaire, send reminders only to stakeholders who have not responded, and stop them when a response is received or the questionnaire closes.
- **FR-DA-COMM-08:** The system shall log every questionnaire dispatch, reminder and response per channel — recipient, channel, timestamp and delivery status — in an immutable record available for audit (Section 4.11).

4.18 Return on Experience (REX)

A Return on Experience (REX) entry captures what a closed Project (or equivalent principal record) or a closed Action actually taught the organization, so the lesson is queryable and reusable rather than lost in an individual's memory or a narrative report.

- **FR-DA-REX-01:** The system shall prompt for a REX entry when the application's principal record reaches a terminal/closed status, and when an individual Action is marked complete, without blocking the closure itself on the REX entry being filled in.
- **FR-DA-REX-02:** The system shall structure every REX entry with, at minimum: what went well, what did not, a root-cause note, a recommendation/improvement action, a category/tag, and an effectiveness or satisfaction rating on a fixed scale.
- **FR-DA-REX-03:** The system shall support full Create, Read, Update, and Delete on REX entries, each optionally tagged to the owning OBS node and to the process/stage it is most relevant to, consistent with how other governance artifacts are tagged (Section 4.6, FR-DA-GOV-08).
- **FR-DA-REX-04:** The system shall index every saved REX entry into the same retrieval corpus used for AI Use Case grounding, the AI Assistant, and application-help search (Section 4.5), scoped to the requesting user's own tenant, so a relevant prior REX entry can surface in a future suggestion or answer.
- **FR-DA-REX-05:** The system shall provide a REX Register view aggregating entries by category/tag and by rating, so a recurring root cause or a declining rating trend is visible without reading every entry individually.
- **FR-DA-REX-06:** The system shall version every REX entry via the generic Version Management service (Section 3.8/4.16), retaining its full edit history and supporting revert to any prior version.
- **FR-DA-REX-07:** The system shall gate REX entry management behind a dedicated capability, and shall scope every REX read and every retrieval-grounded reference to it to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-REX-08:** The system shall present the REX prompt to the user who performed the closing action, immediately after the closure succeeds, with a choice to fill it in now or skip; skipping shall leave the closure intact and the entry available to add later from the REX Register.

4.19 Navigation Shell & Personalization

The navigation shell (Section 3.11) behaves the same way in every Dynamic App, so a user who has learned one application can find their way around any other.

- **FR-DA-NAV-01:** The system shall generate the primary menu from a single navigation model in which each item declares its module group, label, icon, route, and the view permission and Solution Pack entitlement it requires, and shall show only the items the current user can open.
- **FR-DA-NAV-02:** The system shall group menu items by module, let the user collapse and expand each group, and mark the current screen as the active item.
- **FR-DA-NAV-03:** The system shall let each user pin the menu open or unpin it. Unpinned, the menu shall collapse to an edge handle, slide over the content when the user points at the edge, focuses the handle, or uses the menu button, and slide away again when the pointer leaves or an item is chosen.
- **FR-DA-NAV-04:** The system shall let each user dock the menu on the left, right, top, or bottom edge of the screen, laying it out vertically on the left and right and horizontally, with drop-down groups, on the top and bottom.
- **FR-DA-NAV-05:** The system shall let each user mark menu items as favorites, shown in a Favorites group at the start of the menu.
- **FR-DA-NAV-06:** The system shall store the dock position, pinned state, collapsed groups, and favorites as server-side user preferences, applied on every sign-in and on any device, and shall offer them both from the menu itself and from the user's Settings screen.
- **FR-DA-NAV-07:** The system shall treat left and right as logical start and end under right-to-left languages, mirroring the dock position and the slide direction when Arabic is selected.
- **FR-DA-NAV-08:** The system shall, below a mobile breakpoint, replace the docked menu with a drawer opened from a menu button and closed by Escape, by an outside tap, or by choosing an item, without changing the user's stored desktop preferences.
- **FR-DA-NAV-09:** The system shall make the shell fully keyboard-operable: a skip-to-content link as the first focusable element, menu and group buttons exposing their expanded state, Escape closing any open menu, and the active item marked as the current page for assistive technologies.
- **FR-DA-NAV-10:** The system shall keep the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point, and user menu in the header bar of every authenticated screen.
- **FR-DA-NAV-11:** The system shall organize the menu by what users do, in a documented order (home and own work, portfolio of records, process design, governance and risk, AI and knowledge, reports, administration, personal settings), so the same capability is always found in the same place across Dynamic Apps.
- **FR-DA-NAV-12:** The system shall show a Search button at the start of the primary menu, in every dock position, pinned or sliding; unpinned, the button stays visible on the edge handle. It shall open the global search (Section 4.37) and also be reachable by a keyboard shortcut.
- **FR-DA-NAV-13:** The system shall let the user filter the menu itself by typing in the search field of the menu, showing the matching items, favorites first, with their module group.
- **FR-DA-NAV-14:** The system shall show a search bar in the header of every authenticated screen, in addition to the Search button of the menu (FR-DA-NAV-12); typing in it opens the global search (Section 4.37) with the text typed.
- **FR-DA-NAV-15:** The system shall give each catalog that users manage — for example functions, process design, document templates and document layout — a direct menu entry, so that its Create, Read, Update and Delete are reachable in one step from the menu.

4.20 Internal Benchmarking

Internal benchmarking (Section 3.12) lets an Organization learn from its own segments and from the other Organizations of its Group. External benchmarking is out of scope.

- **FR-DA-BMK-01:** The system shall define its benchmark metrics in one catalog module, each with a label, a unit, a direction of good, a plain-language explanation, and a computation from the platform's own records, consumed by every benchmarking screen.
- **FR-DA-BMK-02:** The system shall compare, within one Organization, every catalog metric across the segments of at least one dimension the application defines (for example record or project type, process track, or owning OBS node), showing each segment's value, the Organization-wide value, and the median.
- **FR-DA-BMK-03:** The system shall compare, across the Organizations of the requester's Group, per-Organization aggregates of the same metrics, and shall rank each Organization per metric according to the metric's direction of good, giving tied values the same rank.
- **FR-DA-BMK-04:** The system shall include in a Group comparison only per-metric aggregates and each contributing Organization's name and sector — never a record, record title, person's name, or identifier from another Organization.
- **FR-DA-BMK-05:** The system shall flag any segment or Organization with fewer records than a minimum sample size (default 3) as not comparable, and shall exclude it from ranks and medians.
- **FR-DA-BMK-06:** The system shall let an Organization Administrator withdraw the Organization from Group benchmarking and rejoin it; while withdrawn, its figures shall not be shown to the other Organizations of the Group. Every change shall be written to the audit log.
- **FR-DA-BMK-07:** The system shall gate internal benchmarking behind a view capability, Group comparison behind a separate capability, and the sharing setting behind a management capability, each enforced server-side.
- **FR-DA-BMK-08:** The system shall refuse any comparison with an Organization outside the requester's Group and shall not import or display external or third-party benchmark data; an Organization with no Group shall see its internal comparison only, with a notice explaining why.
- **FR-DA-BMK-09:** The system shall render each metric comparison as a chart following the common visual identity, with the requester's own Organization or the selected segment highlighted, and shall let the user filter the comparison by the application's segment dimensions.

4.21 Platform Operations

These requirements keep a Dynamic App safe to upgrade, recover, and demonstrate, whatever its business domain (Section 3.13).

- **FR-DA-OPS-01:** The system shall take a consistent backup of its data store on demand — from the command line and from the platform administrator's configuration screen — and automatically once a day, while remaining online.
- **FR-DA-OPS-02:** The system shall keep backups for a configurable retention period (default 14 days), delete older ones automatically, and list existing backups with date and size to the platform administrator only.
- **FR-DA-OPS-03:** The system shall document how to restore a backup, and a restored backup from an earlier release shall start without any manual migration step (FR-DA-OPS-04).
- **FR-DA-OPS-04:** The system shall apply additive schema and permission-catalog migrations automatically on startup, never dropping or rewriting existing data or administrator changes, so an existing installation upgrades in place.

- **FR-DA-OPS-05:** The system shall check, when installed and started, the runtime version, port availability, and whether the data store has been initialized, and shall report each problem as a plain-language message naming the corrective action rather than a stack trace.
- **FR-DA-OPS-06:** The system shall expose an unauthenticated health endpoint reporting only liveness, whether the data store is initialized, and the deployment mode — never tenant data.
- **FR-DA-OPS-07:** The system shall ship a deterministic demonstration seed, reset by one command, containing at least one Group of two or more Organizations, at least one Independent Organization, several sectors, a user for every role, and at least ten instances of each principal end-to-end process per Organization, all fictional (Section 2.7).
- **FR-DA-OPS-08:** The system shall include an administrator traceability view mapping each requirement ID of this standard to its implementation status (Met, Partial, Not met, or Deployment responsibility) and a short evidence note.
- **FR-DA-OPS-09:** The system shall seed demonstration data in every module, so that no screen of the delivered application is empty: record teams with roles, templates, attachments in several formats, custom AI Use Cases and project overrides, alert settings and alerts already read, and integration and webhook registrations.

4.22 Portfolio Overview

The portfolio overview answers "where does each record stand in each process?" for any scope the user may see.

- **FR-DA-PFO-01:** The system shall let the user choose a scope: a Group, one or more Organizations, or a set of principal records, each list containing only what the user may see (FR-DA-TEN-15).
- **FR-DA-PFO-02:** The system shall show one row per principal record and one column per process stage, with at each intersection the status of that stage for that record, drawn from one status vocabulary (for example Completed, In progress, At gate, On hold, Stopped, Not started, Not in the record's variant).
- **FR-DA-PFO-03:** The system shall show a legend of the cell statuses, a per-stage total, the number of times a stage ran when more than once, and a hover text naming record, stage and status.
- **FR-DA-PFO-04:** The system shall export the matrix to CSV (FR-DA-REP-04) and shall show records of another Organization read-only.

4.23 Checklist Library

The Checklist Library holds every checklist as an independent entry. A checklist may be related to a gate or not: when it is associated with a gate of the Gate Library (Section 4.35) it becomes part of that gate's exit criteria; when it is not, it can still be attached to a phase, a task or a project and used on its own.

- **FR-DA-CHK-01:** The system shall keep a Checklist Library with full Create, Read, Update, Delete and Duplicate on checklist templates, each with ordered items marked mandatory and evidence-required, classified by scope (universal or one vertical) and optionally by mode and track, and versioned under the generic Version Management service (Section 3.8). A checklist shall not require a gate.
- **FR-DA-CHK-02:** The system shall let a checklist template be related to zero, one or several gates; the association is made from the gate (FR-DA-GTE-02) or from the checklist, and a checklist associated with a gate shall be copied into that gate's checklist when the phase opens. A built-in reference checklist shall apply to a gate that has no associated checklist.

- **FR-DA-CHK-03:** The system shall let an authorized user add items to an open gate or phase checklist from any template or by typing them, never adding the same item twice, and save such a checklist as a new template.
- **FR-DA-CHK-04:** The system shall show for every checklist the gates it is associated with, and let the library be filtered by gate, by "not related to a gate", by vertical, mode and track; it shall seed sector checklists for new Organizations of that sector.
- **FR-DA-CHK-05:** The system shall freeze a checklist, its items and their evidence once the gate it belongs to has been submitted for decision or, for a checklist not related to a gate, once it has been signed off.
- **FR-DA-CHK-06:** The system shall let a checklist that is not related to any gate be attached directly to a phase, a task or a project, with its completion tracked and reported; such a checklist shall not block any progression.
- **FR-DA-CHK-07:** The system shall show, before a checklist is changed or deleted, the gates, templates and open phases that use it; a checklist associated with a published gate shall be retired rather than deleted, and deleting a checklist shall never alter the copies already made in projects.

4.24 Attachments

Actions and checklist items carry the documents that support them.

- **FR-DA-ATT-01:** The system shall let a user attach several files at once to an action or a checklist item, by file picker or drag-and-drop.
- **FR-DA-ATT-02:** The system shall accept a published list of formats — documents, spreadsheets, presentations, images, drawings and CAD, data and models, archives, audio and video, e-mails — with a size and count limit per upload, and shall refuse any other type with a message naming the refused file.
- **FR-DA-ATT-03:** The system shall show for every attachment its type, size, author and date, with download for every viewer and removal for its author or a manager, audited, and never once the stage is submitted.
- **FR-DA-ATT-04:** The system shall store attachments outside every web-served directory, scoped to the tenant, and include them in the backup instructions (NFR-DA-SEC-13, FR-DA-OPS-01).
- **FR-DA-ATT-05:** The system shall version attachments: a user may upload a new version of an attachment with a note; the versions form one chain with number, author and date, the latest is shown by default, and every earlier version stays downloadable.
- **FR-DA-ATT-06:** The system shall let attachments be added to steps, documents and registers as well as to actions and checklist items, under FR-DA-ATT-01 to 05.

4.25 Vertical Framework & Governance

A vertical is a first-class, governed configuration entity. Its content is data, so any Dynamic App can serve new sectors without code changes.

- **FR-DA-VRT-01:** The system shall support the definition of verticals, each with a unique identifier, a short prefix, a name, a description, its target segments, its value proposition, and its associated macro and E2E processes.
- **FR-DA-VRT-02:** The system shall support a taxonomy of verticals with parent-child and sibling relationships.
- **FR-DA-VRT-03:** The system shall let administrators and partners define new verticals without modifying the platform, and new verticals shall work alongside existing ones.

- **FR-DA-VRT-04:** The system shall manage each vertical through the lifecycle Draft, Review, Approved, Active, Deprecated and Retired, with an approval workflow (configurable approvers, escalation and notifications) for definition, activation, modification, deprecation and retirement.
- **FR-DA-VRT-05:** The system shall version-control vertical definitions under the Version Management service (Section 3.8), so that every change can be tracked, reviewed, approved and rolled back, including the rollback of an activation or configuration change to a previous version.
- **FR-DA-VRT-06:** The system shall let each tenant activate its own set of verticals without affecting other tenants.
- **FR-DA-VRT-07:** The system shall refuse to deactivate a vertical, a macro process or an E2E process while active projects depend on it, shall require approval for a deactivation, and shall preserve all data of a deactivated vertical.
- **FR-DA-VRT-08:** The system shall keep an immutable, searchable and exportable audit trail of every vertical action — definition, activation, modification, deactivation and retirement — retained per policy (Section 4.11).
- **FR-DA-VRT-09:** The system shall record a designated owner for every vertical and shall support change management for it: change request, impact analysis, approval and tracked implementation.
- **FR-DA-VRT-10:** The system shall capture and report governance metrics per vertical: adoption, compliance and performance.

4.26 Vertical Processes

Selecting a vertical activates its own processes, built on the common process model of the application.

- **FR-DA-VPR-01:** The system shall activate the macro processes of a vertical when the vertical is activated, with a configurable range of three to five per vertical.
- **FR-DA-VPR-02:** The system shall describe every vertical macro process with an identifier, name, description, goals, input suppliers, inputs, tasks, outputs, output customers, related E2E processes, related modules and applicable compliance requirements.
- **FR-DA-VPR-03:** The system shall map each vertical macro process to the regulatory and industry standards that apply to it; the mapping shall be configurable and audited.
- **FR-DA-VPR-04:** The system shall let administrators and partners add vertical macro processes without modifying the platform, versioned and reversible like verticals (FR-DA-VRT-05).
- **FR-DA-VPR-05:** The system shall support tasks within each macro process (owner, deadline, dependencies, status) and steps within each task (owner, inputs, outputs, status), and shall enforce the dependencies defined between macro processes, tasks and steps.
- **FR-DA-VPR-06:** The system shall activate the E2E processes of a vertical when the vertical is activated, with a configurable range of two to four per vertical.
- **FR-DA-VPR-07:** The system shall enforce the chain relationships between E2E processes — sequential, branching, parallel, loop-back and terminal — and shall support, per E2E process, triggers (business event, time-based, manual) and terminal states (success, failure, cancellation, loop-back).
- **FR-DA-VPR-08:** The system shall support user-facing tasks in each E2E process, each linked to the macro processes, tasks and steps it relies on.
- **FR-DA-VPR-09:** The system shall capture metrics for each vertical macro and E2E process — cycle time, completion rate, quality and compliance — and produce status, compliance and performance reports from them (Section 4.8).

4.27 Vertical Data, Compliance & Integration

Each vertical may extend the data model, carry its own compliance obligations and connect to sector systems, always without breaking the common core.

- **FR-DA-VDT-01:** The system shall let a vertical extend the common data model with its own objects, attributes, relationships, validation rules and lifecycle states, without altering or breaking the common entities (Section 6).
- **FR-DA-VDT-02:** The system shall exchange vertical data in industry-standard formats, at least JSON, XML, CSV and PDF, and, where the vertical needs it, engineering formats such as STEP, IGES and JT.
- **FR-DA-VDT-03:** The system shall apply per-vertical retention policies and support archiving and retrieval of vertical data, in line with the regulatory retention of the sector.
- **FR-DA-VCP-01:** The system shall support vertical-specific regulations and standards as configurable, audited and reportable compliance requirements, subject to FR-DA-CFG-09.
- **FR-DA-VCP-02:** The system shall monitor compliance per vertical, alert stakeholders of non-compliance through the Notification Center (Section 4.9), and produce compliance reports, audit reports and certification records.
- **FR-DA-VCP-03:** The system shall support internal and external compliance audits — planning, execution, findings and corrective actions — and vertical compliance training with content, delivery, tracking and certification.
- **FR-DA-VIN-01:** The system shall connect vertical-specific external systems through standard interfaces (REST, SOAP, GraphQL, file-based, message-based), under the integration rules of FR-DA-CFG-10 to 14.
- **FR-DA-VIN-02:** The system shall secure, monitor (performance, errors, data quality) and test every vertical integration, and shall handle integration errors by retry, fallback and notification.

4.28 Vertical Configuration

Vertical parameters are configured, validated and versioned before activation.

- **FR-DA-VCF-01:** The system shall let an administrator configure, per vertical: macro and E2E process activation, checklist templates, gate criteria and data model extensions.
- **FR-DA-VCF-02:** The system shall provide a pre-built configuration template per vertical for rapid deployment.
- **FR-DA-VCF-03:** The system shall validate a vertical configuration for completeness, consistency and compliance before activation, and shall refuse an activation that fails validation, naming each failed rule.
- **FR-DA-VCF-04:** The system shall version every vertical configuration change and audit who changed what, when and why.

4.29 SME Mode & Tracks

The SME mode is a cross-sector enabler: an SME in any vertical can run it alongside that vertical.

- **FR-DA-SME-01:** The system shall define an SME by configurable employee thresholds per region (default 1–250 in the EU, up to 500 under the US SBA definition) and shall segment SMEs into Micro (1–10), Small (10–50) and Medium (50–250).
- **FR-DA-SME-02:** The system shall let an Organization activate the SME mode together with any vertical, and each tenant shall activate its own tracks and configuration without affecting other tenants.
- **FR-DA-SME-03:** The system shall support two to four configurable SME tracks, each defining its number of macro processes, E2E processes, gates, checklist items per gate, and typical duration.

- **FR-DA-SME-04:** The system shall activate three to five SME macro processes and two to six SME E2E processes when a track is selected, with the same taxonomy, task, step, chain and metric rules as vertical processes (FR-DA-VPR-02, 05, 07, 08 and 09) plus track applicability and cost metrics.
- **FR-DA-SME-05:** The system shall let administrators define new SME tracks, macro processes and configurations without modifying the platform.
- **FR-DA-SME-06:** The system shall assign a project in SME mode to the track recommended by its complexity score (Section 4.30); a manual override shall require a written justification and an approval, and shall be audited and reportable.
- **FR-DA-SME-07:** The system shall capture and report, per track, its usage, effectiveness and outcomes.
- **FR-DA-SME-08:** The system shall govern the SME framework like verticals: a designated owner, an approval workflow for track definition, activation, modification, deprecation and retirement, change management, an immutable audit trail and adoption, compliance and performance metrics.

4.30 Complexity Scoring

A complexity score gives every project the right amount of rigor, consistently and traceably.

- **FR-DA-SCO-01:** The system shall compute a complexity score for every project from weighted criteria and shall use it to recommend the mode and track.
- **FR-DA-SCO-02:** The system shall let an administrator configure the criteria — at least strategic impact, technical novelty, regulatory burden, investment size, cross-functional scope, market uncertainty and time-to-market pressure — and their weights, which shall sum to 100 %; a set of weights that does not sum to 100 % shall be refused.
- **FR-DA-SCO-03:** The system shall let each vertical add its own complexity drivers (for example regulatory burden, industry standards, sector-specific risks) with their weights, versioned, so that the score and the recommended track reflect the vertical of the project.
- **FR-DA-SCO-04:** The system shall keep an audit trail of every score — criteria, weights, values, result and track assignment — and shall report the distribution of scores, track assignments and outcomes.

4.31 SME Onboarding & Experience

SMEs need to go live quickly with little PLM experience.

- **FR-DA-ONB-01:** The system shall run an onboarding plan for an SME within a configurable period (default 30 days): tenant provisioning (FR-DA-TEN-14), track configuration, data import, validation, user training and go-live.
- **FR-DA-ONB-02:** The system shall import data from spreadsheets, CSV files, legacy systems and cloud storage, preserving data integrity and relationships, and shall validate the imported data for completeness, consistency and accuracy before go-live, listing every rejected record with its reason.
- **FR-DA-ONB-03:** The system shall provide self-service training and guided onboarding: interactive tutorials, guided workflows, sample data and contextual help (Section 4.12).
- **FR-DA-ONB-04:** The system shall offer SME users simplified navigation, a mobile-responsive or progressive web app experience, and self-service support through the knowledge base, the AI Assistant (Section 4.10) and a community space.
- **FR-DA-ONB-05:** The system shall measure onboarding: time-to-value, data quality, user adoption and satisfaction.

4.32 Packaging & Pricing

SME packs, vertical packs and their bundles extend the Solution Pack model of Section 4.3.

- **FR-DA-PKG-01:** The system shall let an administrator define three to five SME packs, each with configurable contents, target segment and price, and bundles of a vertical pack with an SME pack, each recorded as a Solution Pack variant so that entitlement stays enforced by FR-DA-CFG-06.
- **FR-DA-PKG-02:** The system shall enforce configurable pricing rules — minimum price per user per month, minimum user count per pack, volume discounts and bundle discounts — and shall show how each price was obtained.
- **FR-DA-PKG-03:** The system shall support upgrading and downgrading between packs and bundles with prorated billing and without loss of data.
- **FR-DA-PKG-04:** The system shall report revenue, discounts, adoption, upgrades and downgrades per pack and bundle, exportable to PDF, XML and CSV.

4.33 Project Template Catalog

Every Dynamic App offers a catalog of project templates. A template is a starting point copied at creation time (FR-DA-TPL-04).

- **FR-DA-PTC-01:** The system shall keep a Project Template Catalog in which every template is classified by scope — Universal (common to all sectors) or one Vertical — and by mode — Full, or SME with its track.
- **FR-DA-PTC-02:** The system shall support full Create, Read, Update, Delete and Duplicate on project templates, versioned under Section 3.8; a template shall carry its default fields, its phases, the gates (Section 4.35) and checklists (Section 4.23) attached to each phase, its OBS roles and its milestones.
- **FR-DA-PTC-03:** The system shall present the catalog as a browsable list with search and filters on scope, vertical, mode, track and status, a preview of each template's phases, gates and checklists, and its use count.
- **FR-DA-PTC-04:** The system shall manage each template through Draft, Published and Retired; only published templates shall be offered at project creation, and retiring a template shall not change any project created from it.
- **FR-DA-PTC-05:** The system shall provide universal templates for Full and SME mode on every installation, seed vertical templates when a vertical is activated, and let an authorized user save an existing project as a new template.
- **FR-DA-PTC-06:** The system shall restrict catalog changes to a dedicated permission, audit every change, and let platform templates be shared read-only with all Organizations while Organization templates stay private to their tenant.

4.34 Project Creation Modes

A project can be created in three ways; all three end in the same, fully editable project.

- **FR-DA-PCM-01:** The system shall offer three creation modes on one entry point: From the catalog, Manual, and With AI.
- **FR-DA-PCM-02:** From the catalog: the system shall propose the templates that match the Organization's verticals and the project's mode and track, and copy the chosen template's fields, phases, gates, checklists and roles into the new project.
- **FR-DA-PCM-03:** Manual: the system shall let the user enter the project and choose its vertical (or none), mode and track, and build its phases by attaching gates from the Gate Library and checklists from the Checklist Library, or by creating new ones.

- **FR-DA-PCM-04:** With AI: the system shall draft the project — fields, complexity scores, recommended mode and track, closest template, phases, gates and checklists — from a short description, as an Assistive AI suggestion that the user accepts, modifies or rejects item by item before anything is saved (FR-DA-AI-07); the draft shall work with the built-in engine and use the live model when configured.
- **FR-DA-PCM-05:** The system shall compute the complexity score (Section 4.30) in every creation mode, apply the quota checks of NFR-DA-SEC-08, and record in the audit trail the creation mode, the template and version used, and the AI suggestions accepted.

4.35 Gate Library

The Gate Library holds every gate definition. Checklists from the Checklist Library (Section 4.23) are associated with a gate; gates are then attached to any phase of any project or project template.

- **FR-DA-GTE-01:** The system shall keep a Gate Library with full Create, Read, Update, Delete and Duplicate on gate definitions, versioned: name, purpose, entry and exit criteria, approvers (roles or people), decision options Go, No-Go, Hold and Recycle, and applicability by vertical, mode and track.
- **FR-DA-GTE-02:** The system shall let an authorized user associate zero, one or several checklists of the Checklist Library with a gate, optionally per track, each marked mandatory or optional, in a chosen order; a checklist created from the gate screen shall be saved in the Checklist Library, and the same checklist may be associated with several gates.
- **FR-DA-GTE-03:** The system shall let an authorized user attach one or more gates to any phase of a project or a project template, reorder them, and detach them while the phase has not started; attaching a gate brings its associated checklists with it.
- **FR-DA-GTE-04:** The system shall configure the gates of a project from its vertical and track: the number of gates depends on the track and the criteria on the vertical, and a gate may be marked optional for a track.
- **FR-DA-GTE-05:** The system shall block the progression of a phase while a mandatory item of a mandatory checklist associated with its gate is incomplete; enforcement shall be configurable per gate and a waiver shall require justification and approval.
- **FR-DA-GTE-06:** The system shall capture every gate decision with its approvers, criteria results and comments, and keep an immutable audit trail of gate actions — creation, modification, checklist association, attachment to a phase, review, decision and waiver.
- **FR-DA-GTE-07:** Changing a gate, or a checklist associated with it, shall not alter the copies already attached to started phases; the user shall be offered to update not-yet-started phases to the new version.

4.36 Questionnaire & Survey Management

A Dynamic App that collects evidence from people — employees, managers, clients or partners — designs questionnaires from a governed template library, distributes them through the channels respondents actually use, and turns consented, complete responses into a dataset its own processes consume.

- **FR-DA-QLT-01:** The system shall support full Create, Read, Update, and Delete on questionnaire templates, each carrying a focus tag (All needs, Digital, AI or a named Custom focus), a sector tag, a language, and a version (FR-DA-TPL-06 to 09).
- **FR-DA-QLT-02:** The system shall let an authorized user elaborate a questionnaire by Load (copying a template), by AI (generating questions from the approved scope through a governed AI use case, Section 4.4), or Manually; the modes may be combined, and the mode of each question shall be recorded.

- **FR-DA-QLT-03:** The system shall label every AI-generated or AI-tailored question and shall block distribution until an authorized user has validated each such question; the validation shall record the validating user and time.
- **FR-DA-QLT-04:** The system shall allocate RACSI roles to each questionnaire section, reusing the RACSI model of Section 4.6, and shall block distribution while any section has no Accountable; a section shall have exactly one Accountable.
- **FR-DA-QLT-05:** The system shall distribute a questionnaire through Face-to-Face, Email, Hybrid and Custom channels, selectable per stakeholder or stakeholder segment, through the questionnaire dispatch abstraction of FR-DA-COMM-06.
- **FR-DA-QLT-06:** The system shall capture the respondent's consent before storing any response; a response submitted without consent shall not be stored, and the rejection shall be logged.
- **FR-DA-QLT-07:** The system shall send reminders on the cadence configured per channel for each questionnaire (FR-DA-COMM-07).
- **FR-DA-QLT-08:** The system shall support offline response collection for the Face-to-Face channel and synchronize captured responses when connectivity returns, without duplicates and with the original capture time preserved.
- **FR-DA-QLT-09:** The system shall compute a completeness percentage for every response and validate it against a configurable threshold before consolidation; a response below the threshold shall be flagged and excluded unless an authorized user includes it with a justification (Section 3.7).
- **FR-DA-QLT-10:** The system shall consolidate validated responses into a structured dataset scoped to the requesting user's tenant, never mixing responses from another Organization.
- **FR-DA-QLT-11:** The system shall make the consolidated dataset available as an input to the application's downstream processes — in CortexSkills, the needs analysis, the diagnostic and the Training Engineering Report — without export or re-keying.
- **FR-DA-QLT-12:** The system shall report response rate, engagement rate and completeness per channel, per focus and per stakeholder segment, exportable under the conventions of Section 4.8.

4.37 Global Search

A user finds anything they may open from one place, whatever the screen they are on.

- **FR-DA-SRCH-01:** The system shall provide a global search, opened from the Search button of the primary menu (FR-DA-NAV-12) or its keyboard shortcut, from every authenticated screen.
- **FR-DA-SRCH-02:** The system shall search, in the current Organization and within the scope of the user's rights: principal records, projects, process elements (functions, processes, phases, tasks, steps, gates, checklists, templates), governance items (rules, controls, risks, KPIs, RACSI), actions, registers, documents, attachment names, AI Use Cases, people and OBS units and roles, and Help topics.
- **FR-DA-SRCH-03:** The system shall match codes and identifiers exactly (for example a step code) and text by words, ignoring case and accents, in every supported language, and shall rank exact code matches first.
- **FR-DA-SRCH-04:** The system shall show results grouped by type, each with its code, title, location (project, process or register) and status, and open the chosen result on its own screen.
- **FR-DA-SRCH-05:** The system shall let the user filter results by type, project, process, status and date, and keep the user's recent searches as a personal preference.
- **FR-DA-SRCH-06:** The system shall apply to every result the RBAC, Solution Pack and tenant rules of the screen it opens (NFR-DA-SEC-18).
- **FR-DA-SRCH-07:** The system shall update the search index when a record is created, changed or deleted, so a change is findable within one minute.

4.38 Process Design Management

The process design is managed as data, with full history, by the people entitled to change it.

- **FR-DA-PDM-01:** The system shall support full Create, Read, Update, Delete and Duplicate on every Process Element: functions, end-to-end processes, phases, macro processes, tasks, steps, gates, checklists and their items, step forms and their fields, templates (project, checklist, document, questionnaire) and AI Use Cases, gated by a process-design management capability.
- **FR-DA-PDM-02:** The system shall version every Process Element under the generic Version Management service (Section 3.8): an edit creates a new version with its author, date, change note and justification (Section 3.7), and no version is rewritten or deleted.
- **FR-DA-PDM-03:** The system shall let an authorized user open any version of a Process Element read-only, compare any two versions field by field (FR-DA-VER-06), and restore any version as the new Current Version.
- **FR-DA-PDM-04:** The system shall version composite elements with their children (FR-DA-VER-08): restoring a version of a macro process restores its tasks and steps, their order, forms, links to gates, checklists, RACSI and AI Use Cases as they were.
- **FR-DA-PDM-05:** The system shall manage Design Releases through Draft, In review, Published and Retired; changes made in a draft release shall not affect running work until the release is published with the approval of an authorized user, and a published release shall be restorable.
- **FR-DA-PDM-06:** The system shall bind every running record to the element versions current when it started, or when its phase opened. A new version applies to new records; an authorized user may migrate running records to it after an impact preview, with a justification, and completed steps shall never change.
- **FR-DA-PDM-07:** The system shall show, before an element is changed, retired or deleted, what uses it: child elements, templates, releases and running records. An element used by a published release or a running record shall be retired rather than deleted; a retired or deleted element shall be restorable from its history.
- **FR-DA-PDM-08:** The system shall let an authorized user reorder elements, move a task or step to another parent and link or unlink elements, with the referential checks of FR-DA-TEN-05; identifiers shall be stable and never reused.
- **FR-DA-PDM-09:** The system shall let the platform share a reference process design read-only with all Organizations, and let an Organization customize it by copy; the copy shall keep the source and version it came from and be offered the changes of a newer source version, element by element.
- **FR-DA-PDM-10:** The system shall export a process design, or part of it, to a portable file (JSON and spreadsheet, with BPMN 2.0 for the diagrams) and import it back as new versions, never overwriting history.
- **FR-DA-PDM-11:** The system shall write every Create, Update, Delete, restore, migration and publication of a Process Element to the audit trail (Section 4.11).

4.39 OBS: Functions, Roles and People

The OBS is made of units, roles and people. A role is linked to one or more functions, and a person can play one or more roles.

- **FR-DA-OBS-01:** The system shall keep a catalog of functions per Organization, with full Create, Read, Update and Delete, versioned, each with a code, a name, a description and an optional parent function, and seed a default set on provisioning (FR-DA-TEN-14).

- **FR-DA-OBS-02:** The system shall support full Create, Read, Update and Delete on OBS Roles, each defined in an OBS unit, with a name, a mission, responsibilities, required competences, an optional reporting role, and the default access roles it proposes to its holders.
- **FR-DA-OBS-03:** The system shall let an OBS Role be linked to one or more functions, and a function to several roles; a role shall be linked to at least one function.
- **FR-DA-OBS-04:** The system shall let a person play one or more OBS Roles, in the same or different units, through Role Assignments with a start date, an optional end date, an allocation percentage and a holder type (holder, deputy, acting), and shall let a role be played by several people.
- **FR-DA-OBS-05:** The system shall show the OBS by unit (units, roles and their holders), by function (the roles and people linked to each function) and by person (the roles a person plays), and as an organization chart.
- **FR-DA-OBS-06:** The system shall let step owners, RACSI letters, approvers, gate approvers, reviewers and notification targets be assigned to OBS Roles, and resolve them to the people currently playing the role; a role with no current holder shall raise an alert (Section 4.9).
- **FR-DA-OBS-07:** The system shall, when a Role Assignment ends, move the open work assigned through that role to its remaining or next holder, and list the work assigned to the person by name for re-assignment.
- **FR-DA-OBS-08:** The system shall offer, wherever a person or a unit is chosen, pickers drawn from the OBS — people with their roles, roles with their holders, units with their type — rather than free text.
- **FR-DA-OBS-09:** The system shall keep OBS Roles distinct from RBAC access roles (Section 3.2); holding an OBS Role shall grant no permission by itself.
- **FR-DA-OBS-10:** The system shall version units, roles, role-function links and Role Assignments under Section 3.8, so the OBS as it was on any past date can be shown, and enforce the quotas of FR-DA-TEN-07.

4.40 AI Use Case Mapping & Prompt Specification

Every AI Use Case is attached to the right task or step and carries a complete, structured and versioned prompt.

- **FR-DA-AIP-01:** The system shall link every AI Use Case to the tasks or steps it supports, the link naming the input fields it reads and the output fields it proposes, and shall show the use case only on those tasks and steps (FR-DA-AI-12, FR-DA-AI-13).
- **FR-DA-AIP-02:** The system shall check each link for fit — the use case's expected output matches the output or a field of the step, and its inputs exist at that point of the process — and flag a link that fails the check for review.
- **FR-DA-AIP-03:** The system shall hold, for every AI Use Case, a Prompt Specification made of separate fields: Role (persona); Context (the process, task or step, the standards and clauses applicable, the Organization profile); Task (the instruction); Inputs (named variables bound to record and step data); Knowledge sources (the corpora and references to retrieve from); Constraints (what the answer must and must not do, including the grounding rule of FR-DA-AI-11); Examples (one or more input and output pairs); Output format (structure, fields, length and a schema where the output fills a form); Tone and language; Quality criteria (how a good answer is judged); Human checkpoint (who reviews and what they check); and Model parameters (model, temperature, maximum length).
- **FR-DA-AIP-04:** The system shall let an authorized user edit each field of a Prompt Specification on its own, and shall version each change under Section 3.8 with its author, date and note.
- **FR-DA-AIP-05:** The system shall let the user see the history of the whole specification and of each field, compare any two versions, and restore any version of one field or of the whole specification as the new Current Version.

- **FR-DA-AIP-06:** The system shall require, before an AI Use Case can be activated, that Role, Context, Task, Constraints, Output format and Human checkpoint are filled, and shall show the completeness of every specification.
- **FR-DA-AIP-07:** The system shall populate every seeded or AI-suggested use case fully for the task or step it is linked to: the Context shall name that task or step, its inputs, outputs and applicable clauses; the Inputs shall bind to its actual fields; the Examples and the Output format shall match its form.
- **FR-DA-AIP-08:** The system shall assemble the prompt from the fields in a fixed, documented order, resolve every variable, and refuse to run when a required variable has no value, naming it.
- **FR-DA-AIP-09:** The system shall let an authorized user test a Prompt Specification on a chosen record before publishing a new version, without writing the result to the record.
- **FR-DA-AIP-10:** The system shall record in the AI Usage Log (FR-DA-AI-07) the use case, the specification version, the engine and model used, and the outcome of every run.
- **FR-DA-AIP-11:** The system shall let an AI Use Case be suggested for a task or step that has none — by rule or with AI — as a draft with a fully populated Prompt Specification, activated only after a person reviews it.
- **FR-DA-AIP-12:** The system shall show with every AI answer the engine that produced it (Engine Transparency) and, when the Organization's live model was not used, a visible message giving the reason — no model enabled, or the error the model returned — with a link to the model settings for users entitled to change them.
- **FR-DA-AIP-13:** The system shall make the built-in engine produce, when it answers, content of the type the use case expects (for example a draft text built from the records already captured), never an echo of the input.
- **FR-DA-AIP-14:** The system shall send as prompt inputs only the values the user typed or the records named in the specification, never the state of interface controls (for example the choice made in a list of outcomes).
- **FR-DA-AIP-15:** The system shall insert an accepted or edited suggestion into the output field the use case names (for example the final validated text), never into an input field.
- **FR-DA-AIP-16:** The system shall let an AI suggestion fill several rows of a record table, each row linked to the records it concerns and marked with the Entry Origin AI, for the user to review, edit or delete before completion.

4.41 Process Element Naming & Descriptions

Names say what is done; descriptions say how.

- **FR-DA-NAM-01:** The system shall require an explicit name for every task and step, beginning with an action verb followed by its object (for example “Identify the interested parties”), unique within its parent, in every supported language.
- **FR-DA-NAM-02:** The system shall check names on save — missing verb, generic wording, abbreviation, duplicate — and show each finding as a warning that the user may accept or correct.
- **FR-DA-NAM-03:** The system shall give every step a brief (one line, shown by default) and a detailed description (purpose, inputs, how to fill each field, expected result), expandable and collapsible, in every supported language.
- **FR-DA-NAM-04:** The system shall show on every step its process, inputs and outputs, the standards and clauses of the record that it covers, and the rules and controls that apply to it.

4.42 Typed Step Forms & Process Readiness

Each step collects the information it needs in a form suited to it, and its completion feeds the registers that follow.

- **FR-DA-SFM-01:** The system shall give every step a Step Form Kind from a catalog — at least record table, assessment matrix, decision, objectives, plan, communication, training, monitoring, review, assignment, configuration, document and free record — each with its own fields, checks and completion effects.
- **FR-DA-SFM-02:** The system shall capture lists as tables of rows, one row per item, with add, edit and delete, each row carrying its source (a named record, study, interview or date) rather than free text.
- **FR-DA-SFM-03:** The system shall provide a decision or assessment matrix with criteria, weights, scores on a stated scale and the facts justifying each score, computing the weighted score and showing the scale next to it (FR-DA-HLP-04).
- **FR-DA-SFM-04:** The system shall, when a step is completed, create or update the records it implies — rows of a plan become actions, objectives become entries of the objectives register, KPI rows become KPI values with their alerts — and link each created record back to its row.
- **FR-DA-SFM-05:** The system shall let a KPI be chosen from the KPI catalog, or created in place, with the reason it is monitored, and let people and units be chosen from the OBS (FR-DA-OBS-08).
- **FR-DA-SFM-06:** The system shall validate the mandatory fields of a form before completion, naming each missing field, and lock the step once the gate of its phase is passed (FR-DA-AUD-05 governs reopening).
- **FR-DA-SFM-07:** The system shall show, before a macro process starts, an optional Readiness Checklist: required inputs, completed predecessor processes, owner and RACSI assigned, templates available and KPIs defined. Items the system can verify shall be checked automatically; the checklist shall inform and never block.

4.43 Documented Information Templates & Generated Documents

Controlled documents and records are produced from the application's own data, not retyped.

- **FR-DA-DOC-01:** The system shall keep a library of documented-information templates with full Create, Read, Update, Delete and Duplicate, versioned, each with a code, a category, the formats it produces (Word, PDF, Excel), and ordered sections typed as text, table or approval block.
- **FR-DA-DOC-02:** The system shall let each section be bound to a named data source of the application (for example context issues, interested parties, objectives, KPIs, risks, RACSI, audits, nonconformities, actions), with free text that may use variables such as the Organization name.
- **FR-DA-DOC-03:** The system shall list, per standard in scope, the documented information it requires — to maintain (a document) or to retain (a record) — with the template and the document that answer each requirement, and create a missing one in one action.
- **FR-DA-DOC-04:** The system shall generate a document from a template and the current data of a project, as a draft version, and let it be regenerated from newer data or edited section by section.
- **FR-DA-DOC-05:** The system shall manage document versions through Draft, In review and Published, with approval by a person other than the author (two-person rule) and a change note; one version is published at a time and earlier versions stay downloadable.
- **FR-DA-DOC-06:** The system shall download every document version in each of its formats, with a cover page, a table of contents where relevant, headers, footers and page numbers.
- **FR-DA-DOC-07:** The system shall let an Organization set its document layout — logo, colors, fonts, header and footer — applied to every generated document.

- **FR-DA-DOC-08:** The system shall let an Organization copy a library template to adapt it; its copy replaces the library template within the tenant and keeps the version it came from.
- **FR-DA-DOC-09:** The system shall let a document be retired, with a justification, rather than deleted once published; a draft may be deleted.
- **FR-DA-DOC-10:** The system shall link generated documents to the step, process and project they come from, and show them on that step.
- **FR-DA-DOC-11:** The system shall seed the template library and, in the demonstration data (FR-DA-OPS-09), documents generated for every seeded project, ready to download.

4.44 Generated Documents: Consistency & Fine-Grained Content

A generated document is detailed, complete for its type and consistent with the application's data and with every other document.

- **FR-DA-DGC-01:** The system shall draw every value of a generated document from the application's records through a Data Source Binding; the same item shall appear with the same identifier, name and values in every screen and every document that shows it.
- **FR-DA-DGC-02:** The system shall trace every data-bound table row and value of a document to its source record (identifier and version), list the data sources used at the end of the document, and show the date and time of the data ("data as of").
- **FR-DA-DGC-03:** The system shall, when the register a section is bound to has no entry yet, use the records typed in the steps that feed it; when no data exists at all, it shall print a visible notice naming the step or register that provides the data, never an empty section or an empty table.
- **FR-DA-DGC-04:** The system shall define a Document Profile per document type, and each generated document shall carry at least: an identification block (reference, title, version, status, owner, approver, dates, classification, distribution), purpose and scope, references and definitions, method and rating scales where scores are used, the data sections of its type, analysis and conclusions, actions with owner and due date, a revision history and an approval block.
- **FR-DA-DGC-05:** The system shall produce data sections at the finest grain of the data: one row per item, each with its own category, description, relevance or score, owner, dates, status and source or evidence, rather than grouped or summarized text.
- **FR-DA-DGC-06:** The system shall, for a procedure or process description, include the process flow as a diagram (BPMN 2.0 with lanes per role, gateways and exception flows), the inputs and outputs of each activity (SIPOC per step), the decision points with their Go / No-Go criteria, evidence and decider, the records produced, the indicators and the risks and controls.
- **FR-DA-DGC-07:** The system shall run Consistency Checks before a document is sent for review and before it is published: referenced items exist and are current; totals, counts and scores agree with their rows; identifiers and names match the records; every mandatory section of the Document Profile is filled; dates are coherent (next review after last review); and the document agrees with the other published documents of the project that show the same items.
- **FR-DA-DGC-08:** The system shall list the findings of the Consistency Checks with their location in the document and a link to the record to correct; findings marked blocking shall prevent publication, the others shall be accepted with a justification (Section 3.7).
- **FR-DA-DGC-09:** The system shall show, for every published document, whether its source data changed since it was generated, list the changed items, and propose a new version; a data change that affects a published document shall raise an alert to its owner (Section 4.9).

- **FR-DA-DGC-10:** The system shall keep the documents of a project mutually consistent: a change of a shared item (for example a party, a risk, an objective, a process owner) shall list every document that shows it and let the owner regenerate them together.
- **FR-DA-DGC-11:** The system shall lay out each format for reading: a cover page and a table of contents where relevant; landscape pages for wide tables; table headers repeated on each page; no heading or table header left alone at the bottom of a page; diagrams rendered as images in Word and PDF; one sheet per table after a cover sheet in Excel; status values shown with the semantic color scale.
- **FR-DA-DGC-12:** The system shall generate a document in any supported language, with the data in that language, and keep the language of each version.
- **FR-DA-DGC-13:** The system shall keep a master list of the project's documented information up to date automatically: reference, title, version, status, owner, approval date, review frequency, next review, retention for records, and the requirement each document answers.

4.45 Generated Documents: Full CRUD at Every Level

Every level of a generated document can be created, read, changed and removed by the people entitled to, with full history.

- **FR-DA-DCR-01:** The system shall support Create, Read, Update, Duplicate and Delete on documents: create from a template or blank, read on screen and in every format, update the metadata (title, owner, approver, review frequency, classification, distribution), duplicate to another project, and delete a document never published; a published document shall be retired with a justification, not deleted (FR-DA-DOC-09).
- **FR-DA-DCR-02:** The system shall support Create, Read, Compare, Restore and Delete on document versions: create a minor or major version by regeneration from newer data or by copy of the current content; read and download any version; compare any two versions section by section and row by row; restore any version as a new draft; delete a draft version only.
- **FR-DA-DCR-03:** The system shall support, on a draft version, Create, Read, Update, Reorder and Delete on sections: add a section (text, data-bound table, diagram, key-value block, approval block), rename it, move it, delete it, edit its text, and change its Data Source Binding, filters and columns.
- **FR-DA-DCR-04:** The system shall let an authorized user edit a row of a data-bound table from the document through Edit-through — add, change or delete the source record under its own rights, checks and audit — and regenerate the section, so the document and the application stay the same.
- **FR-DA-DCR-05:** The system shall let an authorized user set a Section Override for text or a value that belongs to the document only; the override shall be marked, kept when the document is regenerated, listed in the Consistency Checks and removable to return to the source data.
- **FR-DA-DCR-06:** The system shall keep, when a document is regenerated, the manual text and overrides of its sections, and show the user what the regeneration changed before the new version is saved.
- **FR-DA-DCR-07:** The system shall apply the same CRUD to templates (FR-DA-DOC-01): sections, their order, text, data sources and Document Profile; a template change shall apply to documents generated after it and be offered to existing drafts.
- **FR-DA-DCR-08:** The system shall support Create, Read, Update and Delete on the document layout of an Organization (FR-DA-DOC-07) and on its per-type variants, with a preview, versioned under Section 3.8.
- **FR-DA-DCR-09:** The system shall gate every document, version, section, row and layout action by its own permission, enforce the two-person rule for approval (FR-DA-DOC-05) and write every action to the audit trail (Section 4.11).

- **FR-DA-DCR-10:** The system shall show where to customize documents from the documents screen and from every document: templates (all documents of a type), layout (all documents of the Organization) and structure (this document only).

4.46 Data Entry Usability

Every form names what it expects, proposes values and gives room to type.

- **FR-DA-DEU-01:** The system shall qualify every field and column title with the object it holds in its context (for example "External issue", "Interested party", "Macro process 1"), never a generic title such as "Item"; a reusable form shall take its titles from the step or record type that uses it.
- **FR-DA-DEU-02:** The system shall present every choice as a Controlled List with a Custom value that opens a free text field, and shall offer again, in the same list, the values already typed in the same table.
- **FR-DA-DEU-03:** The system shall let the user open any row of a record table in a Row Editor — one large field per column, scrolling, previous and next row, and a return to the table — in edit mode and in read-only mode; a new row shall open in the Row Editor.
- **FR-DA-DEU-04:** The system shall let reference items (processes, phases, functions, people, units, records) be chosen from pickers of the current project or Organization, shown as "ID (name)", with a Custom value where an item outside the application is allowed.
- **FR-DA-DEU-05:** The system shall explain every computed or scored field under the field — what it measures, how it is computed and where it is used — and every free-text field that feeds a document.
- **FR-DA-DEU-06:** The system shall keep the user's unsaved input when a dialog or Row Editor is closed with Back, and ask for confirmation only when the input would be lost.

4.47 Linked Records, Choice Libraries & Register Propagation

Records are captured once, linked to each other and reused everywhere.

- **FR-DA-LNK-01:** The system shall support Record Links many to many between records (for example a need or expectation and the interested parties it concerns, a risk and the processes it affects), each end chosen from the records already identified in earlier steps or registers, or typed as a Custom value.
- **FR-DA-LNK-02:** The system shall let multi-valued records be entered in three ways — typed (Manual), taken from a Choice Library (Library) or suggested by the AI (AI) — and record the Entry Origin of each.
- **FR-DA-LNK-03:** The system shall build each Choice Library from the values entered before in the Organization, in any project, together with a reference list managed as configuration data; library values are not linked to any record until the user links them.
- **FR-DA-LNK-04:** The system shall apply Register Propagation when a step is completed: each row of an identification step creates or updates the matching register entry, matched by its identifier or its name, and keeps the identifier of that entry; linked records update the entries they concern.
- **FR-DA-LNK-05:** The system shall keep steps, registers, pickers and documents consistent: an item typed in a step is offered in the pickers of later steps, shown in its register, and printed in the documents bound to that register (FR-DA-DGC-03).

4.48 Audits, Inspections & Graded Findings

Where a Dynamic App plans audits or inspections, it plans them at a stated frequency and grades each finding.

- **FR-DA-AFP-01:** The system shall let the frequency of an audit or inspection be chosen from a list (for example monthly, quarterly, semi-annual, annual, every two years, every three years) or set as Custom with its description, and derive the next planned date from it.
- **FR-DA-AFP-02:** The system shall record, for each planned audit, its objectives, criteria, scope, processes, duration, lead and team with their qualification, and justify its frequency from the risks and the results of earlier audits.
- **FR-DA-AFP-03:** The system shall grade every finding on a configurable scale — at least major nonconformity, minor nonconformity, observation and opportunity for improvement — with the grading rules shown to the user.
- **FR-DA-AFP-04:** The system shall hold a detailed record for each nonconformity: requirement and clause, statement, objective evidence, process and auditee, root cause, correction, corrective action with owner and due date, verification of effectiveness and status, linked to the corrective action process of the application.
- **FR-DA-AFP-05:** The system shall produce the audit report from these records: summary, strengths, findings by grade and by requirement, one detailed sheet per nonconformity, conclusion and follow-up (Section 4.44).

5. Common Non-Functional Requirements

5.1 Performance

- **NFR-DA-PERF-01:** Dashboard and list views shall render within 2 seconds under nominal per-tenant data volumes.
- **NFR-DA-PERF-02:** An AI agent suggestion shall return within 5 seconds under its deterministic path.
- **NFR-DA-PERF-03:** A knowledge-base or FAQ retrieval search shall return ranked results within 1 second for a tenant's full corpus.
- **NFR-DA-PERF-04:** A single standard-report export (PDF, Excel, or Word) shall complete within 3 seconds for a tenant's current data volume.
- **NFR-DA-PERF-05:** With 100 concurrent users on one tenant, the 95th-percentile response time of read endpoints shall stay at or below 1 second, measured by a repeatable load test. A per-user read cache, where used, shall hold an entry for no more than 30 seconds and shall be invalidated by any write in the same Organization.
- **NFR-DA-PERF-06:** Activating a vertical, with its processes and configuration, shall complete within 5 minutes (configurable target).
- **NFR-DA-PERF-07:** A bulk data import shall load 10,000 records within 30 minutes, and an SME shall be able to complete onboarding within 30 days (configurable targets).
- **NFR-DA-PERF-08:** A production deployment offering the Vertical and SME layers shall be sized for configurable targets, by default 1,000 API calls per second sustained and 10,000 concurrent users, with reports generated within 60 seconds.
- **NFR-DA-PERF-09:** The global search shall show its first suggestions within 300 milliseconds of typing and full results within 1 second at the 95th percentile, for a tenant's full data volume.
- **NFR-DA-PERF-10:** Restoring a version of a process element with up to 1,000 descendants shall complete within 5 seconds; generating a document of up to 50 pages from a template shall complete within 10 seconds.
- **NFR-DA-PERF-11:** Generating a document of up to 50 pages with diagrams (for example BPMN flows) shall complete within 15 seconds, and the consistency checks of a project's documents within 30 seconds, at the 95th percentile.

5.2 Security

- **NFR-DA-SEC-01:** All API endpoints except login shall require a valid, signed, time-boxed authentication token.
- **NFR-DA-SEC-02:** Passwords shall be stored only as salted cryptographic hashes; plaintext passwords shall never be logged or persisted.
- **NFR-DA-SEC-03:** Every authorization decision shall be enforced server-side; client-side permission and Pack-entitlement checks are a UX convenience only, never the security boundary.
- **NFR-DA-SEC-04:** No data query shall be constructed by concatenating untrusted input; all data access shall use parameterized statements or an equivalent safe-query mechanism.
- **NFR-DA-SEC-05:** Owner/Evaluator segregation of duties, and any equivalent server-enforced business rule, shall be enforced server-side, not merely hidden in the UI.
- **NFR-DA-SEC-06:** A one-Accountable-per-RACSI-activity rule, and any equivalent uniqueness rule, shall be enforced by a data-persistence-layer constraint, not merely by disabling an option in the UI.

- **NFR-DA-SEC-07:** A Real LLM Provider Connection's API key shall be captured and stored only client-side; the backend shall never persist it to any table, write it to any log line, or return it in any API response, and any server-side proxying of a forwarded key shall use it for exactly one outbound call and never write, log, cache, or hold it beyond that request.
- **NFR-DA-SEC-08:** Every quota check (Projects, OBS nodes, AI Use Cases, and any other Pack-bound dimension) shall be enforced at the point of creation/activation on the server, never solely by disabling a client-side control; an over-quota request shall be rejected with HTTP 409, never silently truncated or partially applied.
- **NFR-DA-SEC-09:** Solution Pack entitlement shall be treated as an authorization axis independent of, and enforced in addition to, RBAC permissions; a route gated by both shall require both checks to pass, each independently testable and independently capable of producing a 403.
- **NFR-DA-SEC-10:** Individually identifiable records requiring restricted visibility (e.g. individual-level readiness or sentiment data) shall be restricted, by RBAC default, to an explicitly named set of roles; other roles shall see only aggregated, de-identified views.
- **NFR-DA-SEC-11:** Tenant data isolation shall be verified by an automated cross-tenant access test suite exercised in the build/release pipeline, not solely by code review.
- **NFR-DA-SEC-12:** The login endpoint shall be rate-limited per client, and every response shall carry standard security headers (content-type sniffing protection, frame protection, and a restrictive cross-origin resource policy).
- **NFR-DA-SEC-13:** Caches, backups, and logs shall never make one tenant's data available to another; backups shall be stored outside every web-served directory and be readable only by the service account and the platform administrator.
- **NFR-DA-SEC-14:** The system shall support standard federated authentication (SAML 2.0, OAuth 2.0, OpenID Connect) and multi-factor authentication.
- **NFR-DA-SEC-15:** Role-based access control shall support scopes by vertical and by SME track in addition to the tenancy scopes.
- **NFR-DA-SEC-16:** Data shall be encrypted at rest (AES-256 or equivalent) and in transit (TLS 1.2 or later), with configurable data residency regions and optional customer-managed keys (bring your own key).
- **NFR-DA-SEC-17:** An Organization-level live-model key (FR-DA-AI-16) and any integration secret stored on the server shall be encrypted at rest with authenticated encryption (for example AES-256-GCM) under a key held outside the database; it shall never be returned in full, logged, cached beyond its request, exported or included in a backup restore to another tenant. NFR-DA-SEC-07 continues to govern a personal connection.
- **NFR-DA-SEC-18:** A search result, count, suggestion or snippet shall never reveal a record, a title or a value that the user could not open on its own screen.
- **NFR-DA-SEC-19:** A generated document shall contain only data the requesting user may read; a document shared or downloaded shall carry its classification and shall not include hidden data (comments, hidden sheets, metadata) beyond what it shows.

5.3 Reliability & Data Integrity

- **NFR-DA-REL-01:** The system shall degrade gracefully if an AI agent or live-LLM call fails: the underlying CRUD operation shall still succeed, with the AI suggestion simply unavailable or served from its deterministic fallback.
- **NFR-DA-REL-02:** The system shall target 99.9% uptime in a production deployment.

- **NFR-DA-REL-03:** A report-export failure shall return a structured error without corrupting or partially writing the underlying report data view.
- **NFR-DA-REL-04:** The persistence layer shall support scheduled backup with a defined retention policy; a production deployment shall not rely on a single unreplicated data file as its sole durability boundary.
- **NFR-DA-REL-05:** The backup schedule shall bound data loss to at most 24 hours, and restoring a backup shall be tested at least once per release.
- **NFR-DA-REL-06:** A production deployment offering the Vertical and SME layers shall support disaster recovery with an RTO under 4 hours and an RPO under 1 hour, automatic failover and automated backups, tightening NFR-DA-REL-05 for that deployment.
- **NFR-DA-REL-07:** Every version of a versioned entity shall be stored complete and checksummed, so that viewing or restoring it reproduces its exact content; a failed restore shall leave the Current Version unchanged.
- **NFR-DA-REL-08:** Generating a document twice from the same data and template version shall produce the same content; every published document version shall keep the data snapshot it was generated from, so it can be reproduced.

5.4 Scalability

- **NFR-DA-SCALE-01:** The architecture shall support horizontal scaling of the web tier independently of the API tier.
- **NFR-DA-SCALE-02:** The system shall support at least 100 concurrent users per tenant without functional degradation.
- **NFR-DA-SCALE-03:** The persistence schema and the retrieval index shall each be swappable for a distributed equivalent (e.g. a managed relational database, a vector database) without changing the API contract.
- **NFR-DA-SCALE-04:** The system shall scale to configurable limits, by default 1,000 tenants per deployment including 10,000 SME tenants in shared mode, 100 million records, 500 macro processes, 200 E2E processes and 100 checklist items per gate.
- **NFR-DA-SCALE-05:** Adding a vertical, an SME track, a template, a gate or a checklist shall require no downtime.

5.5 Usability & Accessibility

- **NFR-DA-UX-01:** The UI, and every generated document, shall follow the suite's common visual identity (color palette, typography, iconography) consistently across all modules and output formats.
- **NFR-DA-UX-02:** The UI shall be responsive and usable on desktop, tablet, and mobile viewport widths, with touch-target sizing meeting WCAG 2.1 AA on mobile.
- **NFR-DA-UX-03:** All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).
- **NFR-DA-UX-04:** Status, severity, and classification fields shall use one consistent color/badge vocabulary across every module, so a user who has learned a meaning in one module recognizes it in every other.
- **NFR-DA-UX-05:** Every interactive target shall measure at least 24×24 CSS pixels (WCAG 2.2 SC 2.5.8), and at least 44×44 CSS pixels on touch screens.
- **NFR-DA-UX-06:** Every release shall pass an automated accessibility scan (e.g. axe-core) of its main screens with no critical or serious violation. Text shall meet WCAG 2.1 AA contrast (4.5:1, or 3:1 for large

text); where a brand color falls short, its darker tone shall be used for text, and any remaining brand conflict shall be recorded in Appendix A.

- **NFR-DA-UX-07:** Every confirmation, warning, and error message shown as a transient notification shall be translated (FR-DA-I18N-07), announced to assistive technologies through a polite live region, and remain readable long enough to act on.
- **NFR-DA-UX-08:** Wide matrices and tables shall scroll inside their own container, keeping the first column visible, so the page never scrolls horizontally on a 390-pixel viewport.
- **NFR-DA-UX-09:** SME users shall complete onboarding within 1 day and need less than 4 hours of training, helped by guided workflows (configurable targets).
- **NFR-DA-UX-10:** User satisfaction shall be measured, with a configurable target (default NPS of 30 or more).
- **NFR-DA-UX-11:** Search shall be fully keyboard-operable (shortcut, arrow keys, Enter, Escape), group its results by type, highlight the matched words, and mirror under right-to-left languages.
- **NFR-DA-UX-12:** Every step shall show a one-line brief with an expandable detailed description, and every form field shall show how to fill it, in the user's display language.
- **NFR-DA-UX-13:** Every dialog shall keep its title and actions in view and scroll its content with a visible scroll bar; read-only text shall be shown in full rather than inside a small scrolling box.
- **NFR-DA-UX-14:** Every identifier shown to a user shall be followed by the name of the item it identifies, in the form "ID (name)" — for example a phase, a process, a step, a function or a record — in lists, pickers and headers.
- **NFR-DA-UX-15:** A computed field shall be read-only, show how it is computed and why it exists, and never be typed.
- **NFR-DA-UX-16:** Text fields shall be sized for their content; a field meant for more than one sentence shall offer at least five lines, in the form or in the Row Editor.

5.6 Localization

- **NFR-DA-I18N-01:** Every layout — navigation, tables, and charts — shall render correctly mirrored under Arabic RTL, not just with mirrored text direction.
- **NFR-DA-I18N-02:** The system shall support configurable languages (default English, French, Arabic, Spanish, German), currencies (default USD, EUR, MAD, ZAR) and regional date, time and number formats.

5.7 Portability / Installability

- **NFR-DA-PORT-01:** Installation shall succeed on a fresh target machine with only a supported Node.js runtime installed — no native C/C++ build toolchain, no required external database server, and no mandatory external API key.
- **NFR-DA-PORT-02:** The same source shall run unmodified across supported operating systems via standard package-manager install/build/start commands.
- **NFR-DA-PORT-03:** Every change shall be built by a continuous-integration pipeline on Linux, Windows, and macOS that installs dependencies, seeds the demonstration data, runs the automated tests — including the cross-tenant suite of NFR-DA-SEC-11 — and builds the web client.
- **NFR-DA-PORT-04:** Each tier shall install, seed, and start with its package manager's standard commands (install, seed, dev or start), documented in a README at the root of the delivery.

5.8 Maintainability / Extensibility

- **NFR-DA-MAINT-01:** Cross-cutting engines — the Action/Task Register, the Alert & Notification Engine, the RBAC layer, the KPI/Reporting engine, and the Audit Trail Service — shall be implemented as composable modules reusable by sibling Dynamic Apps via the same API contract.
- **NFR-DA-MAINT-02:** AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an LLM/vector-DB-backed one without changing calling code.
- **NFR-DA-MAINT-03:** The Solution Pack feature and quota matrix shall be defined in exactly one place and consumed by every enforcing route, so entitlements can never drift between routes.
- **NFR-DA-MAINT-04:** The AI Use Case Library's generation service shall be the single designated server-side integration point for a live LLM call and its provider-specific credential/endpoint logic; no other module shall embed its own LLM provider client.
- **NFR-DA-MAINT-05:** The alert type-to-name and type-to-stage/process mapping shall be defined in exactly one shared module, consumed by both seed data and live computation.
- **NFR-DA-MAINT-06:** The Installation Guide and User Guide shall be generated from source in Word and PDF, each with an automatic table of contents, and the User Guide's step-by-step walkthroughs shall be replayed automatically against a freshly seeded instance, so the documentation cannot drift from the application.
- **NFR-DA-MAINT-07:** Every User Guide walkthrough shall start from the tenancy — create or choose the Group (Yes/No), create the Organization, then the principal record(s) — and the guide shall include the decision matrix of FR-DA-HLP-04.
- **NFR-DA-MAINT-08:** Verticals, SME tracks, complexity criteria, packs, project templates, gates and checklists shall be configuration data loaded without code changes, and the platform shall be modular with zero-downtime upgrades, monitoring, alerting and comprehensive logging.
- **NFR-DA-MAINT-09:** The process design, functions, OBS roles, Prompt Specifications, step form kinds and documented-information templates shall be configuration data, changed without code changes or downtime.
- **NFR-DA-MAINT-10:** Controlled Lists, Document Profiles, step-specific field titles and data source bindings shall be configuration data, changed without code changes or downtime.

5.9 Compatibility

- **NFR-DA-COMPAT-01:** The application shall target evergreen desktop, tablet, and mobile browsers; no legacy-browser support is in scope.
- **NFR-DA-COMPAT-02:** BPMN diagrams shall be stored and exchanged as schema-valid BPMN 2.0 XML, verified by a round-trip import/export, so a diagram authored in one Dynamic App opens without error in any other BPMN 2.0-compliant tool, and vice versa.
- **NFR-DA-COMPAT-03:** The application's core workflows (record CRUD, approvals, alerts) shall remain fully usable on a mobile viewport without requiring a separate native application; a native or installable Progressive Web App wrapper, where offered, shall consume the same REST API (Section 7.2) with no mobile-specific endpoint variant.

5.10 Auditability & Compliance

- **NFR-DA-COMP-01:** Every AI-generated suggestion shall be logged with a confidence indicator and be attributable to a specific agent/use case and record, or, for the AI Assistant, to the requesting user and question.
- **NFR-DA-COMP-02:** The Controls module shall support demonstrating COSO Internal Control — Integrated Framework coverage across all five components.

- **NFR-DA-COMP-03:** The system shall retain the full version history of every AI Use Case and every template indefinitely (no hard deletion of prior versions on edit or revert).
- **NFR-DA-COMP-04:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **NFR-DA-COMP-05:** The system shall let verticals map to sector standards — for example IATF 16949 (automotive), AS9100 (aerospace), EU MDR and ISO 13485 (medical devices), FDA 21 CFR Part 11 (pharmaceutical), HACCP (food safety) — as scaffolds subject to FR-DA-CFG-09; certification (ISO/IEC 27001, SOC 2, GDPR) is an attribute of a deployment, not of the software.
- **NFR-DA-COMP-06:** Every requirement of a Dynamic App shall be traceable to its business requirement, user story, test case, release and, where applicable, customer commitment, in line with ISO/IEC/IEEE 29148; requirement changes shall be classified Minor, Moderate, Major or Breaking with the matching approval and regression scope.

5.11 Data Retention & Privacy

- **NFR-DA-DATA-01:** By default, and whenever no Real LLM Provider Connection is configured, the system shall not transmit tenant data to any third-party service; every AI capability shall run entirely within the deployment's own infrastructure using rule-based/statistical methods and RAG grounding. Only an explicitly user-configured, client-stored Real LLM Provider Connection may cause a single generation request to be sent to that user-selected third-party provider — opt-in per user, never enabled by default, and never applied to a permission-scoped data-query mode.
- **NFR-DA-DATA-02:** The system shall support both shared-infrastructure (SaaS) and dedicated/sovereign deployment options per Organization, without changing the tenant-isolation guarantee.

5.12 Questionnaire Data

- **NFR-DA-QLT-01:** A questionnaire response shall be stored within 2 seconds of submission, at the reference load of Section 5.1.
- **NFR-DA-QLT-02:** Questionnaire response data shall be retained according to the tenant's retention policy, by default 5 years, and then deleted or anonymized.
- **NFR-DA-QLT-03:** For a questionnaire marked anonymized, the system shall strip the respondent's identity before storage, so that no stored response can be linked back to a stakeholder.

6. Common Data Model

The following core entities recur, under equivalent names, across the Dynamic Apps Suite. Each entity carries a tenant (organization_id) foreign key except Group and any global, platform-wide catalog.

Entity	Description
Group	Optional holding entity above Organization; groups sibling Organizations.
Organization	The tenant boundary; holds sector/country/profile attributes and all tenant-isolated data.
Project	Optional sub-entity of an Organization; the principal record type may reference one; count enforced against the active Pack's quota.
ObsNode	A node in the OBS tree; referenced by users and by every governance entity below; count enforced against the active Pack's quota.
User / Role / Permission	Person, named role, and a global permission catalog entry (module.action) granted to roles.
License / Configuration	Configuration Management record: Solution Pack, Add-On flags, Compliance Standard flags, deployment option, support tier, seats, billing.
GovernanceSettings	KPI thresholds, alert toggles, and process-default/justification-requirement settings per Organization.
Standard / StandardRequirement	A reference standard and its clause-level requirement breakdown, each illustrated with an example.
Principal Record (e.g. NCPFiche / CM Project)	The central aggregate the application manages through its defined multi-stage lifecycle.
Action / ActionEvaluation / ActionEvidence	Abstract action base, its Owner/Evaluator-segregated effectiveness verdict, and supporting evidence attachments.
NotificationAlert	An instance of a cataloged alert type, tagged with its process/stage.
AIAgentLog / AIUsageLog	Immutable logs of every AI agent invocation and every AI suggestion's accepted/edited/rejected outcome.
AuditLog	Immutable log of every create/update/delete on a governed entity.
AIUseCase / AIUseCaseVersion / AIUseCaseProjectOverride	Governed AI Use Case metadata, its versioned content, and its per-Project tri-state activation override.
BusinessRule / Control / RiskOpportunity	Typed governance records, each optionally tagged to an OBS node, a process/stage, and (Control) a Compliance Standard.
RacsiActivity / RacsiAssignment	A governed activity and its R/A/C/S/I assignments, at most one Accountable per activity.
BpmnDiagram	A stored BPMN 2.0 process diagram (XML payload), owned by an OBS node.
CustomKpi	An organization-defined KPI additive to the platform's built-in KPI set.
Template (Sheet/Charter/Communication/Training, etc.)	A versioned, pre-filled starting point for the application's principal record type or a related artifact.
WbsNode	A node in a Work Breakdown Structure, referencing one or more Action IDs as leaf-level work items, with planned dates, percent-complete, and dependency links to other WBS nodes/Actions.
EntityVersion	A generic, immutable version record produced by the Version Management service for any entity type registered with it; carries the entity's full state at that version, the acting user, and a timestamp.
CommunicationDispatch /	A composed message routed through one or more channels (in-app, email, SMS,

Entity	Description
DeliveryStatus	push, webhook) and its per-channel delivery status (queued/sent/delivered/failed).
RexEntry	A Return on Experience record captured at the closure of a principal record or an Action: what went well, what didn't, root cause, recommendation, category/tag, and an effectiveness rating; versioned and indexed into the retrieval corpus.
ExternalIntegration	A registered connection to an external application (CRM, ERP, calendar, storage, identity provider, etc.), its own enable/disable state, Solution-Pack/Add-On gate, credentials reference, and connection-health/audit log.
UserPreference	Per-user settings stored server-side: display language, menu dock position, pinned state, collapsed menu groups, favorites, and notification channel choices.
BenchmarkMetric / BenchmarkSharing	A catalog entry of a comparable metric (unit, direction of good, computation), and an Organization's Group-sharing setting, audited on every change.
Backup	A dated, consistent copy of the data store, with its size and retention expiry; listed to the platform administrator only.
RequirementTrace	A requirement ID of this standard with its implementation status and evidence note, shown in the traceability view.
ObsMember	A person placed in an OBS node, with the standard role and the role played there.
ChecklistTemplate	A versioned list of checklist items in the Checklist Library, universal or specific to a vertical, mode or track, associated with zero or more gates (GateChecklist).
Attachment	A file attached to an action or a checklist item: name, type, size, author, date, tenant-scoped storage reference.
Vertical	A sector entity: prefix, name, taxonomy parent, lifecycle state, owner, version; links to its macro and E2E processes, data extensions, compliance mappings and configuration.
VerticalActivation	The activation of a vertical by an Organization, with its version, validation result and activation order.
SmeTrack	An SME rigor level: processes, gates, checklist items per gate, typical duration, lifecycle state.
ComplexityCriterion	A scoring criterion and its weight, universal or specific to one vertical, versioned.
ComplexityScore	The scored criteria of a project, the result, the recommended and chosen track, and any override justification and approval.
ProjectTemplate	A catalog entry: scope (universal or vertical), mode and track, status, version, fields, phases with their attached gates and checklists, roles and milestones.
GateDefinition	A Gate Library entry: criteria, approvers, decision options, applicability, version.
GateChecklist	The association of a checklist template with a gate, optionally per track, with its order and whether it is mandatory.
PhaseAttachment	The attachment of a gate, or of a checklist not related to a gate, at a given version, to a phase of a project or a template.
Pack	An SME pack, vertical pack or bundle: contents, target segment, pricing rules.
OnboardingPlan	The onboarding of an SME: steps, imports, validation results, dates, metrics.
QuestionnaireTemplate	A versioned questionnaire template — sections and questions — tagged by focus, sector and language; seeded or organization-authored.
Questionnaire	A questionnaire instance loaded, generated or written for an engagement, with its elaboration mode per question, AI-validation state, RACSI allocation per section, status and due date.
QuestionnaireDistribution	The dispatch of a questionnaire to one stakeholder through one channel, with its reminders and per-channel delivery status.

Entity	Description
QuestionnaireResponse	One respondent's answers, with channel used, consent, capture and submission times, completeness percentage and validation; stripped of identity when the questionnaire is anonymized.
Stakeholder	A registered respondent: role, function, decision level, site, preferred channel and consent status.
Function	A business function of an Organization; versioned; linked to OBS Roles and process elements.
OrgRole	An OBS Role defined in an OBS unit, with mission, responsibilities and proposed default access roles.
RoleFunction	The link between an OBS Role and a Function (many to many).
RoleAssignment	A person playing an OBS Role: start and end dates, allocation, holder type.
ProcessElement / ProcessElementVersion	Any element of the process design and its immutable versions.
DesignRelease	A named snapshot of chosen element versions, with its lifecycle status.
RecordDesignBinding	The element versions a running record, or one of its phases, is bound to.
PromptSpecification / PromptFieldVersion	The structured prompt of an AI Use Case and the versions of each field.
AIUseCaseLink	The link of an AI Use Case to a task or step, with the input and output fields it uses.
StepForm / StepFormField	The form kind of a step and its fields, checks and completion effects.
ReadinessItem	An item of the readiness checklist of a macro process, with its automatic check.
DocumentTemplate / TemplateSection	A documented-information template and its typed, data-bound sections.
Document / DocumentVersion	A controlled document of a project and its versions, with status, approval and content.
DocumentLayout	The logo, colors, fonts, header and footer of an Organization's documents.
AttachmentVersion	A version of an attachment in its chain, with number, author, date and note.
SearchIndexEntry	A tenant-scoped, rights-aware index entry of a searchable item.
DocumentProfile	The minimum content and checks of a document type.
DocumentSection	A section of a document version: kind, order, text, Data Source Binding, filters and columns.
DataSourceBinding	The link of a section to a named data source, with filters and sort order.
SectionOverride	Text or a value typed in a document section that replaces the source data for that document.
DocumentSourceTrace	The source record (identifier and version) of each data-bound value of a document version.
ConsistencyFinding	A finding of a Consistency Check: location, rule, severity, status, justification.
ChoiceList / ChoiceValue	A Controlled List and its values, reference or typed by users, with language labels.
RecordLink	A many-to-many link between two records, with its type.
EntryOrigin	The origin of a record entered in a table: Manual, Library or AI (attribute of the record).
ModelProfile	The parameters a language model accepts, its reasoning behavior and its output limits.

Entity	Description
AuditPlan	A planned audit or inspection with frequency (list or custom), objectives, criteria, scope and team.
Finding	A graded audit or inspection finding and, for a nonconformity, its detailed record.

7. Common External Interface Conventions

7.1 User Interface Conventions

Every screen is accessible only when the authenticated user holds the corresponding view permission and, where applicable, the required Solution Pack entitlement. A Dashboard, a Configuration Management screen, a Permission Matrix screen, a Help screen, an Alerts inbox, a Reports screen with per-format download actions, and an AI Assistant (dedicated screen plus floating widget) are present in every Dynamic App.

Every screen sits inside the common navigation shell (Section 4.19). A Benchmarking screen, with Within the organization and Across the group views, is present in every Dynamic App that holds comparable records (Section 4.20), and a Backups panel is present on the platform administrator's configuration screen (Section 4.21). A screen the user may not open shows an access notice (FR-DA-RBAC-08) rather than an empty page.

Revision 1.2 adds a tenancy screen (Group Yes/No > Organization > records), a portfolio overview, a checklist template library in the process design, an attachments panel on every action, AI badges on the process design, and a live AI model card in personal settings and in administration.

Revision 1.3 adds screens for verticals and their activation, SME tracks and complexity criteria, packs, the project template catalog, a project creation entry with three tabs (From the catalog, Manual, With AI), a Checklist Library (checklists related to a gate or not) and a Gate Library (gates with their associated checklists), both attachable to phases.

Revision 1.5 adds a Search button at the start of the primary menu with a global search screen; a process design workspace with version history, comparison and restore on every element; OBS views by unit, by function and by person with an organization chart; a Prompt Specification editor with one field per aspect; typed step forms with record tables and decision matrices; a readiness checklist on every macro process; and a Documents screen with required documents, templates and layout.

Revision 1.6 adds a search bar in the header of every screen; direct menu entries for managed catalogs (functions, document templates and layout); a Row Editor on every record table; Controlled Lists with a Custom value; "ID (name)" labels; scrollable dialogs with fixed title and actions; a Consistency Check panel, a version comparison and section editing on documents; and an engine notice on every AI answer.

7.2 Software Interfaces (REST API)

All interfaces are exposed as a JSON REST API, secured by a bearer token (except the login endpoint), and enforced per-endpoint by the RBAC permission catalog and, where applicable, Solution Pack entitlement middleware. At minimum, every Dynamic App exposes equivalents of the following endpoint families:

- Auth: login, current-user profile, language preference update.
- Hierarchy & OBS: Group/Organization/Project CRUD, OBS CRUD, quota-checked creation.
- Users, Roles & Permissions: user CRUD, role assignment, permission-matrix read/write.
- Configuration Management: read effective configuration and static catalog (open to any authenticated user); write Pack/Add-On/Compliance selection (permission-gated); read live usage against quota.

- Governance: Business Rules, Controls, Risks & Opportunities, RACSI, Custom KPIs — each with process/stage filtering.
- AI Use Cases: catalog CRUD, activation, per-Project override, version history and revert, append-only usage log.
- BPMN: diagram CRUD, view requiring at least view-mode entitlement, write requiring full-edit entitlement.
- AI Assistant: a data-query endpoint and an application-help endpoint, each Pack-gated.
- Reports: per-report data endpoints and a per-report export endpoint accepting a target format (pdf/xlsx/docx).
- Alerts: list, run/recompute, mark-read.
- User preferences: read and update the current user's navigation and display preferences.
- Workflow steps: complete, and reopen with justification (FR-DA-AUD-05).
- Benchmarking: metric catalog, within-organization comparison, Group comparison, and read/update of the Organization's sharing setting.
- Operations: health (unauthenticated, no tenant data); backup list and on-demand backup (platform administrator).
- Tenancy and portfolio: tenancy tree, scope lists (groups, organizations, records the user may see), records across the scope, portfolio matrix.
- Checklist templates: library CRUD with version history; add to a gate or phase checklist from a template or by hand; save it as a template.
- Attachments: accepted types, multi-file upload, download, delete.
- AI model: model and provider catalog; connection test.
- Verticals: CRUD, lifecycle transitions and approvals, versions and rollback, activation with validation, deactivation with dependency check.
- SME and scoring: tracks CRUD, complexity criteria and weights, score computation, track override with approval.
- Project templates: catalog list with filters, CRUD, duplicate, publish and retire, version history, save a project as a template.
- Project creation: create from a template, manually, or from an AI draft (draft, then accept with the retained items).
- Checklist Library: CRUD, duplicate and versions; filter by gate or "not related to a gate"; gates using a checklist; attach a stand-alone checklist to a phase, task or project.
- Gate Library: CRUD, duplicate and versions; associate, order and detach checklists; attach, reorder and detach gates on a phase; decisions and waivers.
- Packs and onboarding: pack and bundle CRUD with pricing rules, upgrade and downgrade; onboarding plan, import and validation.

7.3 Hardware Interfaces

None. A Dynamic App is a standard web application with no direct hardware interface requirement.

7.4 Communications Interfaces

HTTP/HTTPS using JSON request/response bodies for all client-server communication, except report-export responses (application/pdf, .xlsx, .docx binary bodies with a Content-Disposition attachment header). Authentication is via a signed bearer token issued at login.

Appendix A — Conformance & Divergence Notes

This standard states the target platform behavior every Dynamic App should converge on. The two source applications compared to produce it are, as of their respective v2.0 SRS documents, not identical in how completely they satisfy it — recorded here for transparency rather than silently smoothed over.

Revision 1.1 adds a third reference, the CortexPLM build of September 2026, and rows for the capabilities this revision introduces. Where a 1.0 target requirement is now observed in CortexPLM, its row says so.

Area	Standard Requirement	Current Divergence
Server-side authorization (NFR-DA-SEC-01/03)	Full JWT authentication and per-endpoint RBAC + Pack-entitlement enforcement, server-side.	RBAC and Pack-gating (Configuration Management module) are enforced client-side only in the current reference build; the API accepts any caller with no credential check. Closing this gap is the single highest-priority item for a production-grade rebuild. Revision 1.1: CortexPLM enforces JWT authentication, per-endpoint RBAC, and Pack entitlement server-side, covered by automated tests.
Persistence model	Any relational or document store that preserves per-entity querying.	NCP Solver uses a normalized relational schema (node:sqlite). journi persists one whole-tenant JSON document per installation and overwrites it wholesale on every save, with no optimistic locking — acceptable for its single-machine reference context, not for concurrent multi-user production use.
Solution Pack model	Three Packs, priced Add-Ons, Compliance Standards as three independent axes.	Both applications implement this model; naming of the three Packs and the specific module list they unlock differs per product (Resolve/Govern/Assure vs. Trailhead/Waypoint/Horizon), which is expected — the axis structure, not the tier names, is the common requirement.
Compliance & Security Standards catalog	A defined set of independently toggleable standards.	NCP Solver ships three (GDPR, ISO/IEC 27001, SOC 2); journi ships seven (adds ISO/IEC 27701, HIPAA, FIPS 140-2, ISO 9001). The mechanism (Section 4.3) is common; the catalog's specific membership is product-specific and may extend beyond this standard's minimum.
AI Assistant chatbot	RBAC-scoped data query mode plus application-help mode, both Pack-gated.	Both applications implement this pattern; journi additionally composes a third, process-knowledge-base retrieval source into its chatbot (Section 4.5), which is a reasonable extension of, not a deviation from, FR-DA-AST-04.
Governance-artifact scoping	Business Rules/Controls/Risks/RACSI taggable to a process or stage.	NCP Solver tags to the S1–S7 Sheet stage directly; journi tags to its Macro Process catalog (a stable, platform-

Area	Standard Requirement	Current Divergence
		wide process taxonomy) rather than a per-project task instance. Both satisfy FR-DA-GOV-08 at the level of granularity each application's own process model supports.
WBS & Gantt Scheduling (Section 4.15)	Any grouping of Actions can be organized into a WBS and rendered as a Gantt chart.	New in this revision. Not present in either Journi v2.0 or NCP Solver v2.0; neither reference application's current build implements Action grouping, scheduling fields, or a Gantt renderer. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Generic Version Management (Section 4.16)	Any registered entity type shares one versioning service with full history and revert-by-duplication.	New in this revision, generalized from the AI Use Case Library and Template Libraries pattern both applications already implement (FR-DA-AI-03, FR-DA-TPL-02). Neither reference application currently exposes this as a shared service other entity types can register with; each still implements versioning per entity type. Revision 1.1: implemented and verified in the CortexPLM reference build.
Multi-Channel Communication (Section 4.17)	A notification is dispatched through one or more channels (in-app, email, SMS, push, webhook) behind a shared interface.	New in this revision. Both reference applications currently deliver notifications in-app only (Section 4.9); neither implements email, SMS, push, or webhook dispatch. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Mobile Compatibility (Sections 2.5, 5.5, 5.9)	Core workflows remain fully usable on a mobile viewport; a PWA/native wrapper, where offered, uses the same REST API.	New in this revision. Both reference applications currently target desktop/tablet only; neither has been verified against a mobile viewport or ships a PWA/native wrapper. Revision 1.1: CortexPLM verified at a 390-pixel viewport with no horizontal scrolling and 44-pixel touch targets; no native wrapper.
Return on Experience / REX (Section 4.18)	A REX entry is captured at closure of a Project or Action, versioned, and indexed for retrieval.	New in this revision. Neither Journi nor NCP Solver currently prompts for or stores a structured lessons-learned entry at closure; any post-mortem practice today is informal and outside the application. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
External Integration Registry (Sections 3.3, 4.3)	An Organization can register, credential, health-check, and independently toggle a connection to an external application.	New in this revision. Both reference applications currently integrate with, at most, one fixed external system each, configured outside the product rather than through a generic, user-facing registry. Recorded here as a target

Area	Standard Requirement	Current Divergence
		requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Navigation Shell & Personalization (Sections 3.11, 4.19)	One shell; menu pinned or sliding, docked on any edge; favorites and collapsed groups stored server-side; RTL mirroring; mobile drawer; keyboard operation.	New in revision 1.1. Implemented and verified in CortexPLM. Neither the journal nor the NCP Solver v2.0 SRS specifies menu personalization; to be assessed.
Internal Benchmarking (Sections 3.12, 4.20)	Metrics compared across segments of one Organization and across the Organizations of one Group, aggregates only, minimum sample, opt-out; no external benchmarking.	New in revision 1.1. Implemented in CortexPLM: 13 metrics by project type, track, and department; Group comparison with ranks, median, opt-out, and a minimum of 3 projects. Not present in journal or NCP Solver v2.0.
Platform Operations (Sections 3.13, 4.21)	Backups with retention, additive startup migrations, startup guards, health endpoint, deterministic seed, traceability view.	New in revision 1.1. Implemented in CortexPLM (daily backup, 14-day retention). Partial for FR-DA-OPS-05: the runtime version is declared in the package manifest and checked at install time, not by the server at startup. Off-site backup copies remain an operator task (Section 2.7).
Opaque identifiers (FR-DA-TEN-11)	Identifiers reveal nothing about other tenants; foreign records answer 404.	New in revision 1.1. Partial in CortexPLM: foreign records answer 404, but record ids are one global sequence, so an id's size can hint at overall platform activity.
Contrast of brand colors (NFR-DA-UX-06)	WCAG 2.1 AA text contrast; darker brand tone for text where needed.	New in revision 1.1. CortexPLM passes the automated scan except two brand-mandated pairs — white text on the primary orange (2.3:1) and orange key figures (3.0:1) — which conflict with the Brand & Graphical Chart Guide and are recorded here rather than changed.
Other 1.1 additions (TEN-09/10, RBAC-08–10, AUD-05, I18N-07/08, VER-06, REX-08, PERF-05, SEC-12/13, REL-05, UX-05/07, PORT-03/04, MAINT-06)	See Appendix B.	New in revision 1.1. All implemented and verified in CortexPLM (load test at 100 users: 95th percentile 0.9 s; CI on Linux, Windows, and macOS).
Tenancy, provisioning and group visibility (FR-DA-TEN-12 – 16)	Tenancy tree and table; organization chosen before a record; provisioning with starting team; read-only group visibility; OBS members with roles.	New in revision 1.2. Implemented and verified in CortexPLM (automated tests: group scope, independent scope, platform scope, provisioning with 24 accounts and duplicate-domain refusal).
Portfolio overview, checklist library, attachments (Sections 4.22 – 4.24)	Scope matrix with legend; stage checklist templates linked or added on demand; multi-format attachments.	New in revision 1.2. Implemented and verified in CortexPLM, including the User Guide walkthroughs replayed against a fresh installation.
AI placement and model selection (FR-DA-AI-12 – 15), BPMN workspace (FR-DA-BPMN-06 – 08)	Tier badges runnable in place; custom use case delete; model list with custom choice and connection test; palette section and full screen.	New in revision 1.2. Implemented in CortexPLM. The connection test was verified for the authentication-failure outcome; a live call needs the user's own key.

Area	Standard Requirement	Current Divergence
Verticals and SME layers (Sections 4.25 – 4.32)	Verticals as governed configuration entities; SME mode with tracks; weighted complexity scoring; onboarding; packs and bundles.	New in revision 1.3. Target. CortexPLM covers part of it: an Organization carries a sector with seeded templates and checklists, and seven fixed criteria recommend a Full, Light or Fast track, with written justification for another choice. Configurable weights, vertical lifecycle and activation, SME tracks, onboarding and packs are not implemented.
Project Template Catalog and creation modes (Sections 4.33 – 4.34)	Catalog classified universal or per vertical, Full or SME; CRUD and versions; creation from the catalog, manually or with AI.	New in revision 1.3. Partly implemented in CortexPLM: project templates with CRUD and versions, creation from a template or manually, and AI suggestions on the project. Scope and mode classification, publication states and a full AI-drafted project are not implemented.
Checklist Library and Gate Library (Sections 4.23 and 4.35)	Checklists as independent entries, related to a gate or not; gates as library entries with CRUD and associated checklists; both attachable to any phase.	Reworded or new in revision 1.3. CortexPLM keeps checklist templates per gate and track, copied when the gate opens, and adds items from a template or by hand; stand-alone checklists, gate CRUD, gate-checklist association and attachment to any phase are a target, because its gates are fixed per track in code.
Questionnaire & Survey Management (Section 4.36; FR-DA-TPL-06 – 09, FR-DA-COMM-06 – 08, NFR-DA-QLT-01 – 03)	Versioned template library by focus and sector; Load / AI / Manual elaboration with human validation; RACSI per section; multi-channel dispatch and reminders; consent-gated, offline-capable collection; tenant-scoped consolidation.	New in revision 1.4, first specified for CortexSkills (change request CR1). Not implemented in journi, NCP Solver or CortexPLM; CortexSkills is the first build to verify it.
Global search (FR-DA-SRCH, NFR-DA-PERF-09)	Search button in the menu; rights-aware search over every item type.	Search button at the top of the menu and Ctrl+K; steps, macro processes, records, register entries, documents, AI Use Cases, people and Help, grouped by type with filters, within the user's rights; menu filter: Met.
Process design management (FR-DA-PDM)	Full CRUD with version history and restore of any version on every process element.	Process design editor with create, read, update, retire or delete and restore on functions, phases, macro processes, tasks, steps, gates and checklists (templates in the document library); every save is a version with compare and restore; usage shown before a change; running projects keep the version they started with. Named Design Releases and import or export of the design are not yet available: Partial (core met).
OBS role model (FR-DA-OBS)	Units, roles linked to functions, people playing several roles.	OBS roles defined in units and linked to one or more functions; people assigned as holder, deputy or acting with allocation and dates; views by role, by function and by person; vacant roles flagged; hand-over of open work

Area	Standard Requirement	Current Divergence
		when an assignment ends; roles versioned and separate from access roles: Met.
AI prompt specification (FR-DA-AIP)	Separate, versioned prompt fields fully populated per task or step.	Each AI Use Case linked to one step; Prompt Specification with one field per aspect (role, context, task, inputs, knowledge, constraints, examples, format, tone, quality criteria, human checkpoint, model parameters), each with its own versions and restore, plus whole-specification versions; completeness required for activation; all use cases populated from their step; assembled prompt visible: Met.
Naming, step forms, readiness, documents (FR-DA-NAM, SFM, DOC)	Explicit names and descriptions; typed forms; readiness checklist; documented-information templates.	Implemented in DynamicMS (September 2026 build), with procedures carrying SIPOC per step, Go / No-Go decisions and BPMN diagrams: Met.
Document consistency and content (FR-DA-DGC)	Detailed, fine-grained documents drawn from the data and checked before publication.	Documents drawn from the project data with identification block, method, one row per item with source, analysis, revision history and approval; procedures with BPMN, SIPOC per step and Go / No-Go; step data used when a register is empty; landscape, anti-orphan and Excel sheets per table. Consistency Checks before publication, change detection on published documents and per-value source trace are not yet available: Partial.
Document full CRUD (FR-DA-DCR)	CRUD on documents, versions, sections, data-bound rows, templates and layout.	Create, read, update, delete (draft) and retire documents; versions draft, review, published, superseded, each downloadable; section add, rename, reorder, delete and text edit on drafts; template and layout CRUD. Version comparison, Edit-through, Section Overrides kept across regeneration are not yet available: Partial.
Data entry usability (FR-DA-DEU, NFR-DA-UX-13 – 16)	Qualified titles, lists with Custom, Row Editor, ID (name).	Qualified column titles per step, Controlled Lists with Custom, Row Editor with previous / next and back, macro process pickers shown as code (name), explained computed fields, scrollable dialogs. The confirmation before losing unsaved input (FR-DA-DEU-06) is not yet available: Met except DEU-06.
Linked records and propagation (FR-DA-LNK)	Many-to-many links, Manual / Library / AI entry, register propagation.	Needs linked to several interested parties; library of needs typed in the Organization plus a reference list; AI suggestions linked to parties; context and interested-party registers filled on completion: Met (demonstrated on the context analysis steps).
Audits and graded findings	Frequency list with Custom; graded,	Frequency list with Custom;

Area	Standard Requirement	Current Divergence
(FR-DA-AFP)	detailed findings.	objectives, criteria, duration, processes; findings graded major, minor, observation, opportunity with a detailed record and report: Met.
AI transparency and model profiles (FR-DA-AIP-12 – 16, FR-DA-AI-19 – 20)	Engine and reason shown; parameters adapted to each model.	Engine and fallback reason shown with a link to the settings; built-in policy draft; only typed inputs sent; suggestion inserted in the output field; sampling parameters omitted for models that reject them; connection test reporting a refused key, a declined request or the model's own error message. Capability profiles are built in for the listed models rather than configurable: Met (profiles Partial).

A future revision of any product's SRS that closes one of the gaps above should update this appendix in the same revision, so the standard and the observed state of the suite never drift silently apart.

Appendix B — Revision 1.1 Change Log

Revision 1.1 is additive. Every 1.0 requirement keeps its ID and wording, except the editorial changes marked Clarified below. New requirements take the next free number in their section, so a conformance record written against 1.0 remains valid against 1.1.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.1; CortexPLM reference	Version raised to 1.1. CortexPLM added as a third reference; purpose and overview explain the revision.
1.2 Scope	Added	Navigation shell, internal benchmarking, and platform operations brought into scope; external benchmarking stated out of scope.
1.4 Definitions	6 terms added	Navigation Shell, Pinned / Sliding Menu, Dock Position, Internal Benchmarking, Minimum Sample Size, Additive Migration.
2.2, 2.3	3 principles, 3 functions	Personal navigation shell; compare within the Group only; upgrade and recover in place.
2.5 Operating Environment	Clarified	Recommended Node.js raised from 22.5+ to 22.13+ (built-in SQLite without an experimental flag); required version declared in the package manifest.
2.6, 2.7	1 constraint, 1 assumption	Benchmark data leaving a tenant is aggregated only; off-site backup copies are the operator's responsibility.
3.1 Tenancy	Clarified	Group membership is the benchmarking boundary; Group Yes/No and Independent shown on hierarchy screens.
3.11 – 3.13	New sections	Navigation Shell & Personalization, Internal Benchmarking, and Platform Operations architectures.
4.1 Tenant & Hierarchy	FR-DA-TEN-09 – 11	Group membership display; same-tree OBS parents; opaque identifiers and 404 for foreign records.
4.2 Identity & RBAC	FR-DA-RBAC-08 – 10	Access notice instead of a blank page; restricted evaluation-content permission; additive permission migration at startup.
4.11 Audit Trail	FR-DA-AUD-05	Reopen a completed step, with justification, before its stage is decided.
4.14 Localization	FR-DA-I18N-07 – 08	Server messages and assistant answers translated; multilingual Knowledge Base seed with language filter.
4.16 Version Management	FR-DA-VER-06	One shared side-by-side version comparison with changed fields highlighted.
4.18 REX	FR-DA-REX-08	REX prompt shown to whoever closes the record, with Skip that keeps the closure.
4.19 Navigation Shell	FR-DA-NAV-01 – 10 (new)	Menu from one navigation model; collapsible groups; pin / slide away; dock left, right, top, or bottom; favorites; server-side preferences; RTL mirroring; mobile drawer; keyboard operation; standard header bar.
4.20 Internal Benchmarking	FR-DA-BMK-01 – 09 (new)	Metric catalog; within-organization segments; Group ranks with ties; aggregates only; minimum sample; opt-out with audit; permissions; no external benchmarking; charts with filters.
4.21 Platform	FR-DA-OPS-01 – 08	On-demand and daily backup; retention; restore;

Section	Requirement IDs / Type	Change
Operations	(new)	additive migrations; startup guards; health endpoint; deterministic seed; traceability view.
5.1 Performance	NFR-DA-PERF-05	95th-percentile read time of 1 s or less at 100 concurrent users; rules for a per-user read cache.
5.2 Security	NFR-DA-SEC-12 – 13	Login rate limiting and security headers; no cross-tenant exposure through caches, backups, or logs.
5.3 Reliability	NFR-DA-REL-05	Data loss bounded to 24 hours; restore tested each release.
5.5 Usability & Accessibility	NFR-DA-UX-05 – 07	Target sizes 24 px / 44 px; automated accessibility scan and contrast rule; accessible, translated notifications.
5.7 Portability	NFR-DA-PORT-03 – 04	CI on Linux, Windows, and macOS with seed, tests, and build; standard install / seed / start commands.
5.8 Maintainability	NFR-DA-MAINT-06	Guides generated in Word and PDF with a TOC; walkthroughs replayed against a fresh seed.
6 Data Model	4 entities	UserPreference, BenchmarkMetric / BenchmarkSharing, Backup, RequirementTrace.
7.1, 7.2 Interfaces	Clarified	Shell, Benchmarking screen, Backups panel, and access notice; endpoint families for preferences, workflow steps, benchmarking, and operations.
Appendix A	Updated	CortexPLM status added to six 1.0 target rows and server-side authorization; six rows added for 1.1 capabilities; closing note widened to any product.

Nothing was removed. Product-specific material from CortexPLM — its phase-gate tracks, gate criteria, industry content, and commercial packaging — was deliberately left out, because it does not apply to every Dynamic App.

Appendix C — Revision 1.2 Change Log

Revision 1.2 is additive, like 1.1: every earlier requirement keeps its ID and wording, and new requirements take the next free number in their section.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.6	Version 1.2	Version raised to 1.2; purpose and overview describe the revision.
1.4 Definitions	4 terms added	Tenancy Tree, Provisioning, Portfolio Overview, Stage Checklist Template.
4.1 Tenant & Hierarchy	FR-DA-TEN-12 – 16	Tenancy tree and table; organization chosen before creating a record; provisioning of new organizations with an optional starting team; read-only visibility within the group; OBS members with roles.
4.4 AI Use Case Library	FR-DA-AI-12 – 15	Tier badges shown and runnable on the process design; links to step, process, task or E2E process; delete for custom use cases; model list with custom choice, custom endpoint and connection test.
4.7 BPMN	FR-DA-BPMN-06 – 08	Shapes palette section (legend when view-only); full-screen workspace with zoom and return; statement of whether diagrams drive execution.
4.12 Help	FR-DA-HLP-04	Decision matrix next to scored inputs; legend for every code or colour vocabulary.
4.19 Navigation	FR-DA-NAV-11	Menu organized by what users do, in a documented order.
4.21 Operations	FR-DA-OPS-09	Demonstration seed covers every module.
4.22 Portfolio Overview	FR-DA-PFO-01 – 04 (new)	Scope, record × stage matrix, legend and totals, CSV export, read-only foreign records.
4.23 Stage Checklist Library	FR-DA-CHK-01 – 05 (new)	Versioned templates per stage and variant; linked templates copied at stage opening; add from template or by hand without duplicates; sector templates; frozen after submission.
4.24 Attachments	FR-DA-ATT-01 – 04 (new)	Multi-file upload; published accepted formats with limits; metadata, download, audited removal; storage and backup.
5.5 Usability	NFR-DA-UX-08	Wide matrices scroll in their own container; no page-level horizontal scroll on phones.
5.8 Maintainability	NFR-DA-MAINT-07	User Guide walkthroughs start from the tenancy and include the decision matrix.
6 Data Model	3 entities	ObsMember, ChecklistTemplate, Attachment.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for tenancy and portfolio, checklist templates, attachments and the AI model.
Appendix A	3 rows	Conformance of the 1.2 additions in CortexPLM.

Product-specific material of this CortexPLM round — its sectors, its phase-gate vocabulary and its demonstration content — stays in the CortexPLM documentation.

Appendix D — Revision 1.3 Change Log

Revision 1.3 keeps every earlier requirement ID. It is additive except in Section 4.23, where five requirements are reworded so that checklists and gates are separate libraries; new requirements take the next free number in their section or a new family.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.3	Version raised to 1.3; purpose and overview describe the revision; the Verticals and SME source added to the references.
1.4 Definitions	11 terms added, 1 reworded	Stage Checklist Template renamed Checklist Template. Added: Vertical, Vertical Instance, SME, Mode (Full / SME), SME Track, Complexity Score, Project Template Catalog, Creation Mode, Phase, Gate Definition, Pack / Bundle.
2.3 Product Functions	1 function added	Verticals, SME mode, complexity scoring, template catalog, creation modes, gates and checklists on any phase.
3.14 (new)	Architecture	Five requirement layers; verticals and tracks as configuration data; ordered, additive activation.
4.13 Template Libraries	FR-DA-TPL-05	Project templates are managed through the catalog of Section 4.33.
4.23 Checklist Library (renamed)	FR-DA-CHK-01 – 05 reworded; CHK-06 – 07 (new)	Renamed from Stage Checklist Library. A checklist is an independent library entry, related to zero, one or several gates; stand-alone checklists attach to a phase, task or project without blocking progression; impact shown before change or deletion.
4.25 Vertical Framework & Governance	FR-DA-VRT-01 – 10 (new)	Definition, taxonomy, extensibility, lifecycle and approval, versioning and rollback, tenant isolation, guarded deactivation, audit, ownership and change management, metrics.
4.26 Vertical Processes	FR-DA-VPR-01 – 09 (new)	Macro processes (3–5) and E2E processes (2–4) per vertical; taxonomy, compliance mapping, tasks and steps, chains, triggers, terminal states, metrics and reports.
4.27 Vertical Data, Compliance & Integration	FR-DA-VDT-01 – 03, VCP-01 – 03, VIN-01 – 02 (new)	Data model extension, exchange formats, retention and archiving; compliance requirements, monitoring, audits and training; connectors, security, monitoring and error handling.
4.28 Vertical Configuration	FR-DA-VCF-01 – 04 (new)	Configurable parameters, pre-built templates, validation before activation, versioning and audit.
4.29 SME Mode & Tracks	FR-DA-SME-01 – 08 (new)	SME definition and segments, cross-sector activation, 2–4 tracks, SME processes, extensibility, assignment and override, metrics, governance.
4.30 Complexity Scoring	FR-DA-SCO-01 – 04 (new)	Weighted configurable criteria summing to 100 %, vertical drivers, audit and reporting.
4.31 SME Onboarding & Experience	FR-DA-ONB-01 – 05 (new)	30-day onboarding plan, import and validation, guided training, simplified mobile experience and self-service support, metrics.
4.32 Packaging & Pricing	FR-DA-PKG-01 – 04 (new)	SME packs and vertical bundles as Solution Pack variants, pricing rules, upgrade and downgrade, reports.
4.33 Project Template	FR-DA-PTC-01 – 06	Universal or vertical, Full or SME templates; CRUD,

Section	Requirement IDs / Type	Change
Catalog	(new)	duplicate and versions; catalog browsing; Draft, Published, Retired; seeding and save-as-template; permissions.
4.34 Project Creation Modes	FR-DA-PCM-01 – 05 (new)	From the catalog, Manual, With AI (reviewed item by item); scoring, quotas and audit in every mode.
4.35 Gate Library	FR-DA-GTE-01 – 07 (new)	Gate definitions with CRUD and versions; association of checklists from the Checklist Library; attachment of gates to any phase; vertical- and track-aware gates; enforcement and waivers; decisions and audit; library changes never alter started phases.
5.1, 5.2, 5.3, 5.4	NFR-DA-PERF-06 – 08, SEC-14 – 16, REL-06, SCALE-04 – 05	Activation, import and production sizing targets; federated authentication and MFA, vertical and track scopes, encryption, residency and BYOK; disaster recovery; scale limits and zero-downtime additions.
5.5, 5.6, 5.8, 5.10	NFR-DA-UX-09 – 10, I18N-02, MAINT-08, COMP-05 – 06	SME onboarding and training targets, satisfaction; languages, currencies and formats; configuration-driven layers; sector standards scaffolds; requirement traceability and change classes.
6 Data Model	11 entities, 1 reworded	Vertical, VerticalActivation, SmeTrack, ComplexityCriterion, ComplexityScore, ProjectTemplate, GateDefinition, GateChecklist, PhaseAttachment, Pack, OnboardingPlan; ChecklistTemplate no longer tied to one stage.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for verticals, SME and scoring, templates, creation modes, gates and checklists, packs and onboarding.
Appendix A	3 rows	Conformance of the 1.3 additions in CortexPLM.
Appendix E (new)	Traceability	Mapping of every source requirement family to the requirements of this standard.

Appendix E — Traceability to the Verticals and SME Requirements

Each requirement family of the Application-Agnostic Requirements for Verticals and SME source is covered as follows. The 29 vertical instances of the source (REQ-VERT-{SECTOR}-...: Automotive, Aerospace & Defense, Industrial Machinery, Healthcare & Life Sciences, Electronics & High-Tech, AEC & Construction, Food & Beverage, Social & Solidarity Economy, Pharmaceutical & Process, Retail, Footwear & Apparel, Consumer Packaged Goods, Energy & Utilities, Telecommunications, Process/Petrochemical, Shipbuilding, Consumer Products Extended, Process Industries, Infrastructure, Fabrication & Assembly, Process/Packaged Goods, Semiconductor & Electronics, Retail, Distribution, Logistics, Pharma, Manufacturing (Generic), Private Education, Public Education, Healthcare Providers) are content delivered as Vertical Instances under Sections 4.25 to 4.28; they stay in the vertical packs, not in this standard.

Source family	Requirements in this standard	Note
REQ-VERT-FW-001 – 010	FR-DA-VRT-01 – 08	FW-005 and FW-008 merged into VRT-04 and VRT-05.
REQ-VERT-GOV-001 – 005	FR-DA-VRT-04, 08, 09, 10	GOV-002 and GOV-004 duplicate FW-005 and FW-010.
REQ-VERT-MP-001 – 010	FR-DA-VPR-01 – 05, 09	Tasks, steps and dependencies merged into VPR-05.
REQ-VERT-E2E-001 – 007	FR-DA-VPR-06 – 09	Triggers and terminal states merged into VPR-07.
REQ-VERT-DATA-001 – 005	FR-DA-VDT-01 – 03	Retention and archiving merged into VDT-03.
REQ-VERT-COMP-001 – 005	FR-DA-VCP-01 – 03, NFR-DA-COMP-05	Kept subject to FR-DA-CFG-09.
REQ-VERT-INT-001 – 005	FR-DA-VIN-01 – 02	Builds on FR-DA-CFG-10 – 14.
REQ-VERT-CONF-001 – 005	FR-DA-VCF-01 – 04	Audit merged into VCF-04.
REQ-SME-FW-001 – 005	FR-DA-SME-01, 02, 05	
REQ-SME-MP-001 – 005, REQ-SME-E2E-001 – 005	FR-DA-SME-04	Reuses the vertical process rules.
REQ-SME-TRACK-001 – 005	FR-DA-SME-03, 06, 07, 08	
REQ-SME-SCORE-001 – 005	FR-DA-SCO-01, 02, 04	
REQ-SME-ONB-001 – 005	FR-DA-ONB-01 – 03, 05	
REQ-SME-PRICE-001 – 005	FR-DA-PKG-01 – 04	Packs are Solution Pack variants.
REQ-SME-UX-001 – 005	FR-DA-ONB-03, 04, NFR-DA-UX-02, 03	WCAG 2.1 AA and mobile use already required in Section 5.5.

Source family	Requirements in this standard	Note
REQ-SME-GOV-001 – 005	FR-DA-SME-08	
REQ-VS-ACT-001 – 003	FR-DA-SME-02, FR-DA-VRT-07, 08, Section 3.14	Vertical processes first, SME processes second.
REQ-VS-SCORE-001 – 003	FR-DA-SCO-03	
REQ-VS-CHK-001 – 003	FR-DA-CHK-01 – 07, FR-DA-GTE-05, 06	Checklists right-sized per vertical and track.
REQ-VS-GATE-001 – 003	FR-DA-GTE-01, 02, 04, 06	
REQ-VS-PKG-001 – 003	FR-DA-PKG-01 – 04	
REQ-NFR-PERF, SCALE	NFR-DA-PERF-06 – 08, NFR-DA-SCALE-04 – 05	Production targets; existing reference targets unchanged.
REQ-NFR-SEC, REL	NFR-DA-SEC-14 – 16, NFR-DA-REL-06	Audit and tenant isolation already covered (Section 4.11, NFR-DA-SEC-11).
REQ-NFR-UX, LOC	NFR-DA-UX-09 – 10, NFR-DA-I18N-02	RTL already covered (NFR-DA-I18N-01).
REQ-NFR-COMP, MNT; Part 7 governance	NFR-DA-COMP-05 – 06, NFR-DA-MAINT-08	

Product-specific material — the content of each vertical instance, the phase-gate vocabulary of each application and its demonstration data — stays in the documentation of that application.

Appendix F — Revision 1.4 Change Log

Revision 1.4 keeps every earlier requirement ID and is purely additive. Change request CR1 named the new section 4.22 and the new template requirements FR-DA-TPL-05 to 08, and asked for a revision 1.2; because revisions 1.2 and 1.3 already use Section 4.22 (Portfolio Overview) and FR-DA-TPL-05 (Project Template Catalog), the new section is 4.36, the new template requirements are FR-DA-TPL-06 to 09, and the revision is 1.4.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.4	Version raised to 1.4; purpose, scope and overview describe the revision; CR1 added to the references.
1.4 Definitions	4 terms added	Questionnaire Template, Focus, Distribution Channel, Stakeholder.
2.3 Product Functions	1 function added	Questionnaire design, multi-channel distribution, consented collection and consolidation.
4.13 Template Libraries	FR-DA-TPL-06 – 09 (new)	Seeded questionnaire template library by focus and sector; organization-authored templates; versioning through Section 4.16; export and re-import.
4.17 Multi-Channel Communication	FR-DA-COMM-06 – 08 (new)	Questionnaire dispatch channel abstraction; reminder cadence per channel and questionnaire; audit log of dispatches, reminders and responses.
4.36 (new) Questionnaire & Survey Management	FR-DA-QLT-01 – 12 (new)	Template CRUD with tags; Load / AI / Manual elaboration; human validation of AI content; RACSI per section; channels; consent; reminders; offline collection; completeness; tenant-scoped consolidation; downstream feed; response reporting.
5.12 (new) Questionnaire Data	NFR-DA-QLT-01 – 03 (new)	Response stored within 2 s; retention per tenant policy, default 5 years; identity stripped for anonymized questionnaires.
6 Data Model	5 entities	QuestionnaireTemplate, Questionnaire, QuestionnaireDistribution, QuestionnaireResponse, Stakeholder.
Appendix A	1 row	Conformance of the 1.4 additions.
Appendix F (new)	Change log	This change log.

Appendix G — Revision 1.5 Change Log

Revision 1.5 keeps every earlier requirement ID and is purely additive. Its requirements are application agnostic: they come from the DynamicMS reference build and from the product direction received with its user feedback, restated so that any Dynamic App can meet them.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.5	Version raised to 1.5; purpose, scope and overview describe the revision; the DynamicMS reference build added to the references.
1.4 Definitions	12 terms added	Function, OBS Role, Role Assignment, Process Design, Process Element, Design Release, Current Version, Prompt Specification, Step Form Kind, Readiness Checklist, Documented Information Template, Global Search.
2.2, 2.3	3 principles, 4 functions	Versioned process design; accountability through roles; explicit prompts. Global search; process design management; OBS role model; generated documents.
3.15 – 3.18 (new)	Architecture	Process design and versioning; OBS role model; AI prompt specification; global search.
4.4 AI Use Case Library	FR-DA-AI-16 – 18 (new)	Organization-level live model stored encrypted; model per use case; view of the assembled prompt and of the engine used.
4.6 Governance	FR-DA-GOV-09 – 10 (new)	RACSI assigned to OBS Roles; RACSI editor per macro process.
4.16 Version Management	FR-DA-VER-07 – 09 (new)	List and restore of any version; composite versions; named baselines.
4.19 Navigation Shell	FR-DA-NAV-12 – 13 (new)	Search button in the menu, in every dock position; menu filter.
4.24 Attachments	FR-DA-ATT-05 – 06 (new)	Attachment version chains; attachments on steps, documents and registers.
4.37 (new) Global Search	FR-DA-SRCH-01 – 07 (new)	Scope, matching, results, filters, rights, index freshness.
4.38 (new) Process Design Management	FR-DA-PDM-01 – 11 (new)	CRUD and versions on every process element; compare and restore; composite versions; Design Releases; binding of running records; impact check; sharing and customization; import and export; audit.
4.39 (new) OBS: Functions, Roles and People	FR-DA-OBS-01 – 10 (new)	Functions; roles linked to one or more functions; people playing one or more roles; views; assignment and resolution through roles; hand-over; pickers; separation from RBAC; history.
4.40 (new) AI Use Case Mapping & Prompt Specification	FR-DA-AIP-01 – 11 (new)	Mapping to tasks and steps with fit check; separate prompt fields; field and whole versions with restore; completeness; full population per step; assembly; test; logging; suggestion of new use cases.
4.41 (new) Naming & Descriptions	FR-DA-NAM-01 – 04 (new)	Verb-object names; naming checks; brief and detailed descriptions; step context.
4.42 (new) Typed Step Forms & Readiness	FR-DA-SFM-01 – 07 (new)	Form kinds; record tables with sources; decision matrix; completion effects; KPI and OBS pickers; validation and lock; readiness checklist.
4.43 (new) Documented	FR-DA-DOC-01 – 11	Template library; data-bound sections; documents

Section	Requirement IDs / Type	Change
Information	(new)	required by the standards; generation; versions and approval; downloads; layout; tenant copies; retirement; links; seed.
5.1, 5.2, 5.3, 5.5, 5.8	NFR-DA-PERF-09 – 10, SEC-17 – 18, REL-07, UX-11 – 12, MAINT-09 (new)	Search and restore times; encrypted Organization key; rights-safe search; complete versions; keyboard search; step descriptions; configuration data.
6 Data Model	16 entities	Function, OrgRole, RoleFunction, RoleAssignment, ProcessElement(Version), DesignRelease, RecordDesignBinding, PromptSpecification, PromptFieldVersion, AIUseCaseLink, StepForm(Field), ReadinessItem, DocumentTemplate, TemplateSection, Document(Version), DocumentLayout, AttachmentVersion, SearchIndexEntry.
7.1 User Interface	1 paragraph	Screens added by revision 1.5.
Appendix A	5 rows	Conformance of the 1.5 additions in the DynamicMS reference build.
Appendix G (new)	Change log	This change log.

Appendix H — Revision 1.6 Change Log

Revision 1.6 keeps every earlier requirement ID and is purely additive. Its requirements are application agnostic: they come from the second and third rounds of DynamicMS user feedback, restated so that any Dynamic App can meet them. The generated-document requirements of Sections 4.44 and 4.45 detail those of Section 4.43.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.6	Version raised to 1.6; purpose, scope and overview describe the revision; the feedback rounds 2 and 3 added to the references.
1.4 Definitions	12 terms added	Document Profile, Data Source Binding, Consistency Check, Section Override, Edit-through, Controlled List, Choice Library, Entry Origin, Record Link, Register Propagation, Row Editor, Engine Transparency.
2.2, 2.3	2 principles, 2 functions	One truth, many documents; usable before complete. Consistent documents with full CRUD; usable record capture with links and propagation.
4.4 AI Use Case Library	FR-DA-AI-19 – 20 (new)	Model capability profiles; plain-language connection test.
4.19 Navigation Shell	FR-DA-NAV-14 – 15 (new)	Search bar in the header of every screen; menu entries for managed catalogs.
4.40 AI Prompt Specification	FR-DA-AIP-12 – 16 (new)	Engine Transparency; useful built-in answers; clean inputs; insertion in the output field; multi-row suggestions with Entry Origin.
4.44 (new) Document Consistency & Content	FR-DA-DGC-01 – 13 (new)	Single source; trace and data date; fallback and notices; Document Profiles; finest grain; procedures with BPMN, SIPOC and Go / No-Go; Consistency Checks and findings; change detection; cross-document consistency; layout; languages; master list.
4.45 (new) Document Full CRUD	FR-DA-DCR-01 – 10 (new)	Documents; versions with compare and restore; sections; Edit-through; Section Overrides; regeneration keeping manual text; templates; layout; permissions and audit; entry points for customization.
4.46 (new) Data Entry Usability	FR-DA-DEU-01 – 06 (new)	Qualified titles; Controlled Lists with Custom; Row Editor; ID (name) pickers; explained fields; unsaved input kept.
4.47 (new) Linked Records & Propagation	FR-DA-LNK-01 – 05 (new)	Many-to-many links; Manual, Library and AI entry; Choice Libraries; Register Propagation; consistency of steps, registers, pickers and documents.
4.48 (new) Audits & Graded Findings	FR-DA-AFP-01 – 05 (new)	Frequency list with Custom; audit plan; grading; detailed nonconformity record; audit report.
5.1, 5.2, 5.3, 5.5, 5.8	NFR-DA-PERF-11, SEC-19, REL-08, UX-13 – 16, MAINT-10 (new)	Document and check times; rights-safe documents; reproducible generation; scrollable dialogs; ID (name); computed fields; field sizes; configuration data.
6 Data Model	12 entities	DocumentProfile, DocumentSection, DataSourceBinding, SectionOverride, DocumentSourceTrace, ConsistencyFinding, ChoiceList / ChoiceValue, RecordLink, EntryOrigin, ModelProfile, AuditPlan, Finding.
7.1 User Interface	1 paragraph	Screens added by revision 1.6.

Section	Requirement IDs / Type	Change
Appendix A	6 rows	Conformance of the 1.6 additions in the DynamicMS reference build.
Appendix H (new)	Change log	This change log.